



YOUR KINDLE NOTES FOR:

Death's End (Remembrance of Earth's Past)

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271 Highlights | 17 Notes

Highlight (Yellow) | Page 19

He had already forgotten about the witch. Unlike his father, Manuel II, and elder brother, John VIII, Constantine was practical and understood that those who put all their faith in miracles tended to meet with untimely ends.

Highlight (Yellow) | Page 21

Her ancestor, a man from the village, had answered their call and sailed across the Mediterranean to fight for God in the Holy Land. He had risen through the ranks and become a Templar Knight. Later, he had come to Constantinople and met a beautiful woman, a holy warrior; they had fallen in love and given birth to this glorious family.... Later, when she was older, she had found out the truth. The basic frame of the story was true: Her ancestor had indeed been a member of the Children's Crusade.

Highlight (Yellow) | Page 21

The Byzantine Empire had been hoping for the elite troops of Christendom to fight off the infidels. Instead, they received an army of frail women as poor as beggars. The Byzantine court refused to supply these "holy warriors," and the women knights became prostitutes. For more than a hundred years, Helena's "glorious" family had barely eked out a living. By her father's time, the family's poverty had grown even more acute. A hungry Helena picked up the trade practiced by her own illustrious ancestor, but when her father found out, he had beaten her, telling her that he would kill her if he ever caught her again ... unless she took her clients back home so that he could negotiate a better price and keep the money "for her."

Highlight (Yellow) | Page 26

The age of magic began at four o'clock on the afternoon of May 3, 1453, when the high-dimensional fragment first intersected with the Earth. It ended at nine o'clock on the evening of May 28, 1453, when the fragment left the Earth behind. After twenty-five days and five hours, the world returned to its normal orbit. On the evening of May 29, Constantinople fell.

Highlight (Yellow) | Page 27

Once, ancient Romans had whistled in their grand, magnificent baths, thinking that their empire, like the granite that made up the walls of the pools in which they floated, would last forever. No banquet was eternal. Everything had an end. Everything.

Highlight (Yellow) | Page 28

Contrary to usual practice, the ring for the collider wasn't an underground tunnel, but enclosed within an aboveground concrete tube. The facility looked like a giant full stop mark in the setting sun.¹ What sentence does it end? Hopefully only the end of physics. Once, Yang Dong had held a basic belief: Life and the world were perhaps ugly, but at the limits of the micro and macro scales, everything was harmonious and beautiful. The world of our everyday life was only froth floating on the perfect ocean of deep reality. But now, it appeared that the everyday world was a beautiful shell: The micro realities it enclosed and the macro realities that enclosed it were far more ugly and chaotic than the shell itself. Too frightening.

Highlight (Yellow) | Page 28

Like many elderly people, Yang's mother wasn't familiar with the details of the web and her own computer, so she had only deleted the decrypted documents instead of digitally shredding them. She didn't realize that even if she had reformatted the hard drive, the data would still have been easily recoverable.

Highlight (Yellow) | Page 29

But between the end of physics and her mother's secret, Yang had lost two such half lives, which added up to a whole life. What did she have left? Yang Dong leaned against the banister and stared at the abyss beneath her, terrified as well as enticed.

Highlight (Yellow) | Page 30

"But the physical parameters governing the existence of life are utterly unforgiving. Take liquid water as an example: It can exist only within a narrow range of temperatures. Viewing the universe as a whole, this becomes even more apparent: If the parameters of the big bang had been different by even one million billionth, we would have no heavy elements and thus no life. Isn't this clear evidence for intelligent design?"

Highlight (Yellow) | Page 31

"What are we looking at?" Yang Dong asked. "Earth. Without life, this is what the surface of the planet would look like now." "But ... where are the oceans?" "There are no oceans. No rivers either. The entire surface is dry." "You're saying that without life, liquid water would not exist on Earth?" "The reality would probably be even more shocking. Remember, this is only a coarse simulation, but at least you can see how much of an impact life had in the present state of the Earth." "But—" "Do you think life is nothing but a fragile, thin, soft shell clinging to the surface of this planet?" "Isn't it?" "Only if you neglect the power of time. If a colony of ants continue to move clods the size of grains of rice, they could remove all of Mount Tai in a billion years. As long as you give it enough time, life is stronger than metal and stone, more powerful than typhoons and volcanoes."

Highlight (Yellow) | Page 31

"Then how did the oceans disappear?" "We'd have to examine the records of the simulation, which would be a lot of work. However, I can give you an educated guess: plants, animals, and bacteria all have had important

roles in the present composition of our atmosphere. Without life, the atmosphere would be very different. It's possible that such an atmosphere would not be able to shield the surface of the Earth against solar winds and ultraviolet rays, and the oceans would evaporate. Soon, greenhouse effects would turn the Earth's atmosphere into a copy of Venus's, and then water vapor would be lost to space over time. After several billion years, the Earth would be dry."

Highlight (Yellow) | Page 32

"Thus, the Earth that we live on now is a home constructed by life for itself. It has nothing to do with God."

Highlight (Yellow) | Page 32

"If we use a similar mathematical model to simulate the entire universe, and uncheck the option for life at the beginning, what would the resulting universe look like?" Green Glasses thought for a moment. "It would look the same. When I talked about the effects of life on the environment, it was limited to the Earth. But if we're talking about the universe, life is exceedingly rare, and its impact on the evolution of the universe can be ignored."

Highlight (Yellow) | Page 32

From her mother's secret documents, she knew that life was not so rare in the universe. In fact, the universe was downright crowded. How much has the universe been changed by life?

Highlight (Yellow) | Page 32

She tried to stop thinking, to turn her mind into empty darkness, but a new question stubbornly refused to leave her alone: Is Nature really natural?

Highlight (Yellow) | Page 33

Tianming read the newspaper and came to the following conclusion: Compared to the time before he was hospitalized, news about Trisolaris and the Earth-Trisolaris Organization (ETO) no longer dominated everything. There were at least some articles that had nothing to do with the crisis. Humanity's tendency to focus on the here and now reasserted itself, and concern for events that would not take place for four centuries gave way to thoughts about life in the present. This wasn't surprising. He tried to remember what was happening four hundred years ago: China was under the Ming Dynasty, and he thought—he wasn't sure—that Nurhaci had just founded the empire that would end up replacing the Ming, after slaughtering millions. The Dark Ages had just ended in the West; the steam engine wouldn't make its appearance for another hundred-plus years; and, as for electricity, one would have to wait three hundred years. If anyone at the time had worried about life four hundred years later, they'd be a laughingstock. It was as ridiculous to worry about the future as to lament the past.

Highlight (Yellow) | Page 36

The man in the white coat then told Lao Li that a question was going to be displayed on the screen, and the same question would be asked five times. Each time the question was displayed, there would be six buttons underneath, numbered from zero to five. If Lao Li wished to answer in the affirmative, he had to click on the specific numbered button indicated in the on-screen instructions, which would change randomly each time the question was asked. If Lao Li wished to answer in the negative, he just had to press zero, and the procedure would stop immediately. There would be no “Yes” or “No” button. The reason for the complicated procedure, the man explained, was to avoid a situation where the patient simply continued to press the same button over and over without thinking about his answers each time.

Highlight (Yellow) | Page 36

Do you wish to terminate your life? For yes, select 3. For no, select 0. Lao Li selected 3. Do you wish to terminate your life? For yes, select 5. For no, select 0. Lao Li selected 5. The process repeated twice more. And then: Do you wish to terminate your life? This is your last prompt. For yes, select 4. For no, select 0. A surge of sorrow made Tianming dizzy, and he almost fainted. Even when his mother died, he didn't feel such extreme pain and anger. He wanted to scream at Lao Li to select 0, to break the glass window, to suffocate that voice. But Lao Li selected 4. Noiselessly, the injector came to life. Tianming could see the column of yellowish liquid in the glass tube shorten and then disappear. Lao Li never moved. He closed his eyes and went to sleep. The crowd around Tianming dissipated, but he remained where he was, his hand pressed against the glass. He wasn't looking at the lifeless body lying within. His eyes were open, but he wasn't looking at all. “There was no pain.” Dr. Zhang's voice was so low that it sounded like the buzzing of a mosquito. Tianming felt a hand land on his left shoulder. “It's a combination of a massive dose of barbitone, muscle relaxant, and potassium chloride. The barbitone takes effect first and puts the patient into a deep sleep, the muscle relaxant stops his breathing, and the potassium chloride stops the heart. The whole process takes no more than twenty, thirty seconds.”

Highlight (Yellow) | Page 39

Don't be too pleased with yourself. She's nice to everyone. Of course Tianming knew that. But then Wen saw the bottle of mineral water in Tianming's hand and the conversation shifted. What in the world are you drinking?! The water in the bottle was a green color, and bits of grass and leaves floated in it. I crumpled some weeds and added them to the water. It's the most organic drink. He was in a good mood and so he was more loquacious than usual. Someday I may start a company to produce this drink. It will surely be popular. It must taste awful. Do you think cigarettes and liquor really taste that good? Even Coca-Cola probably tasted medicinal the first time you tried it. Anything addictive is like that. “Buddy, that conversation changed my life!” Wen said. He opened the cardboard box and took out a can. The outside was deep green, and on it was a picture of a grassland. The trademark was “Green Tempest.”

Highlight (Yellow) | Page 41

A physician can only cure diseases meant to be cured; the Buddha can only save those meant to be saved.”

Highlight (Yellow) | Page 43

“Crisis Infantilism.” It was commonly thought that Infantilism was a response to an unprecedented threat to the entirety of civilization. That might have been true for individuals, but it was too simple an explanation when applied to humanity as a whole. The Trisolar Crisis’s impact on society was far deeper than people had imagined at first. To give some imperfect analogies: In terms of biology, it was equivalent to the moment when the ancestors of mammals climbed from the ocean onto land; in terms of religion, it was akin to when Adam and Eve were banished from Eden; in terms of history and sociology ... there are no suitable analogies, even imperfect ones. Compared to the Trisolar Crisis, everything heretofore experienced by human civilization was nothing. The Crisis shook the very foundation of culture, politics, religion, and economics. Although the impact reached the deepest core of civilization, its influence manifested most quickly at the surface. The root cause for Crisis Infantilism may well be found in the interaction between these manifestations and the tremendous inertia of human society’s inherent conservatism. The classic examples of Crisis Infantilism were the Wallfacer Project and the Stars Our Destination Project, both international efforts within the framework of the United Nations—initiatives

Highlight (Yellow) | Page 48

She would have preferred a ceremony to generate as much media coverage as possible, but she refrained out of respect for his wish for privacy. She seemed quite sorry to lose out on a wonderful opportunity to publicize the project. Don’t worry, Tianming thought. No other Chinese will be as dumb as me.

Highlight (Yellow) | Page 52

He had been born to an intellectual family, but his parents lacked political savvy and social cunning, and they had not been successful in their lives.

Highlight (Yellow) | Page 52

He was only permitted to read classic books and listen to classical music; the friends he tried to make had to be the kind that his parents deemed to be from cultured, refined families. They told Tianming that the people around them were vulgar, their concerns common. In contrast, their own tastes were far superior.

Highlight (Yellow) | Page 62

the first words out of Wade’s mouth froze her face and entire body. “Would you sell your mother to a whorehouse?”

Highlight (Yellow) | Page 62

After she told Vadimov what had happened, Vadimov laughed. “That’s just a line that used to be popular in our ... trade. I heard it started back during the Second World War. Veterans would use it as a joke on novices. The point is: Our profession is the only one on Earth where lies and betrayal are at the very heart of what we do. We have to be ... flexible when it comes to commonly accepted ethical norms. PIA is formed from two groups

of people: Some are technical experts like you; others are veterans of the various intelligence agencies in the world. These two groups have different ways of thinking and acting. It's a good thing that I'm familiar with both and can help you adjust to the other." "But our enemy is Trisolaris. This is nothing like traditional intelligence." "Some things are constant."

Highlight (Yellow) | Page 62

It was just like Vadimov had predicted: These people were far more interested in spying on each other than gathering intelligence on Trisolaris.

Highlight (Yellow) | Page 63

"The PIA must launch a spy probe at the Trisolaran Fleet." The stunned staff members looked at each other.

Highlight (Yellow) | Page 63

this is something that they won't compromise on: The probe must achieve one percent of lightspeed.

Highlight (Yellow) | Page 64

Wade shook his head. "With those sophons zipping about at lightspeed, spying on us constantly, and completely blocking all fundamental physics research, it's no longer certain that we'll make significant technological progress in the future. If humanity is doomed to crawl at a snail's pace through space, we'd better get started as soon as possible."

Highlight (Yellow) | Page 64

"That is precisely why the probe must achieve one percent of lightspeed." "You're talking about boosting our current maximum speed a hundredfold! That requires a brand-new form of propulsion."

Highlight (Yellow) | Page 64

Wade slammed his fist down on the table. "You forget that we now have resources! Before, spaceflight was merely a luxury, but now it's an absolute necessity. We can ask for resources that far exceed what was imaginable before. We can throw resources at this problem until the laws of physics bend. Rely on brute force if you have to, but we must accelerate the probe to one percent of lightspeed!"

Highlight (Yellow) and Note | Page 69

twenty-five square kilometers,

9.7 sq mi

Highlight (Yellow) | Page 71

As long as the person could be safely and successfully inserted into the fleet, the possibilities were endless. Given that the Trisolarians were transparent in thought and vulnerable to stratagems, Wade's idea became even more attractive. We need to send a representative of humanity into the heart of the enemy.

Highlight (Yellow) | Page 72

Hibernation: Man Walks for the First Time Through Time

Highlight (Yellow) | Page 72

A new technology can transform society, but when the technology is in its infancy, very few people can see its full potential. For example, when the computer was first invented, it was merely a tool for increasing efficiency, and some thought five computers would be enough for the entire world. Artificial hibernation was the same. Before it was a reality, people just thought it would provide an opportunity for patients with terminal illnesses to seek a cure in the future. If they thought further, it would appear to be useful for interstellar voyages. But as soon as it became real, if one examined it through the lens of sociology, one could see that it would completely change the face of human civilization. All this was based on a single idea: Tomorrow will be better.

Highlight (Yellow) | Page 72

after the Industrial Revolution, progress became a constant feature of society, and humanity's faith in the future grew stronger.

Highlight (Yellow) | Page 74

The entire system—hibernation tank, life-support system, cooling equipment—weighed about three metric tons. As Cheng Xin discussed possible ways to miniaturize the hibernation setup with the center's technical staff, she was startled by a realization: If the body's temperature must be maintained around minus-fifty-degrees Celsius, then in the frigid conditions of outer space, the hibernation chamber needed to be heated, not cooled. In the long journey through trans-Neptunian space in particular, outside temperature would be close to absolute zero.

Highlight (Yellow) | Page 75

"Why do you think your countries sent you here? To collect a paycheck and to give me bad news? I don't have a solution. Finding a solution is your job!"

Highlight (Yellow) | Page 76

Cheng Xin wasn't paying much attention to this discussion. Instead, she was focused on one thing: Who would this minus-two-hundred-degree corpsicle that would be shot into deep space be?

Highlight (Yellow) | Page 79

Wade's figure appeared as a black ghost, his eyes glinting with a cold light. "We'll send only a brain," he said.

Highlight (Yellow) | Page 80

Both of these weapon systems were attempts to utilize primitive technology in novel ways that demonstrated a power incongruous for their time period. Looking back, the Staircase Program implemented at the beginning of the Crisis Era was a similar advance. Using only the primitive technology available at the time, it managed to boost a small probe to 1 percent of lightspeed. This achievement should have been impossible without technology that would not appear for another one and a half centuries.

Highlight (Yellow) | Page 81

The cables were made from a nanomaterial called "Flying Blade." Only about a tenth of the thickness of a strand of spider silk, the cables were invisible to the naked eye. Eight grams of the material could be stretched into a cable one hundred kilometers long, yet it was strong enough to securely pull the payload capsule during acceleration, and would not break from the massive radiation generated by the nuclear explosions.

Highlight (Yellow) | Page 82

He argued that if we assumed the Trisolarans were capable of reviving a body in deep freeze, then it made sense to assume they were also capable of reviving a bare brain in similar condition and conversing with it through an external interface. Surely such a task was trivial for a civilization capable of unfolding a proton into two dimensions and etching circuits over the resulting surface.

Highlight (Yellow) | Page 87

Tianming was terrified of space. Like everyone who studied spaceflight for a living, he understood space's sinister nature better than the general public. Hell was not on Earth, but in heaven. Cheng Xin wanted a part of him, the part that carried his soul, to wander forever in that frigid, endless, lightless abyss.

Highlight (Yellow) | Page 92

She had recommended him because she was trying to protect the Earth and its civilization, and his life had almost reached its end—had she not arrived in time, he would be dead. In a way, she had saved him! She had done nothing that she ought to be ashamed of, nothing that should trouble her conscience. But she also understood that this was how someone could sell their mother to a whorehouse.

Highlight (Yellow) | Page 97

she recognized the white-haired old man: the world's foremost brain surgeon, affiliated with Harvard Medical School. He had been the lead surgeon during the operation. "It would be the body that had carried this brain in the first place. Every cell in the brain contains all the genetic information necessary to reconstruct his body. They could clone him and implant the brain, and in this way, he would be whole again."

Highlight (Yellow) | Page 100

At least the Staircase Program was a half success. For the first time, a man-made object had been accelerated to quasi-relativistic speeds.

Highlight (Yellow) | Page 105

Electromagnetic waves are, moreover, the most primitive and least power-efficient method of transmitting information in the universe. Radio waves attenuate and degrade rapidly in the vastness of space, and most electromagnetic signals spilling from the Earth could not be received beyond two light-years. Only something like the transmission by Ye Wenjie, which relied on the power of the sun as an antenna, could be intercepted by listeners among the stars. As humanity's technology advanced, two far more efficient methods of signaling became available: neutrinos and gravitational waves. The latter was the main method of deterrence that humanity later deployed against Trisolaris. The dark forest theory had a profound impact on human civilization. That child sitting by the ashes of the campfire turned from optimism to isolation and paranoia, a loner in the universe.

Highlight (Yellow) | Page 107

The warship stopped accelerating and repeatedly queried Earth for confirmation. No reply ever came, as Earth maintained radio silence. Just as Bronze Age was about to begin accelerating away from home again, the unimaginable happened: A sophon unfolded into low dimensions on the ship, establishing a quantum communication channel with Earth. Finally, the crew received confirmation of all that had occurred.

Highlight (Yellow) | Page 108

But Gravity was an exception. It was a white cylinder, so regular that it seemed unreal, like a basic shape dropped into space by mathematical modeling software, a platonic ideal rather than reality. If the crew of Bronze Age had seen the gravitational wave antennas on the Earth, they would have recognized Gravity as an almost perfect replica of them. Indeed, the entire hull of Gravity was a large gravitational wave antenna. Like its twins on the Earth's surface, the ship was capable of broadcasting gravitational wave messages toward all corners of the universe at a moment's notice. These gravitational wave antennas on Earth and in space comprised humanity's dark forest deterrence system against Trisolaris.

Highlight (Yellow) | Page 114

The Germans made a film based on Jones's experiment, and Jones himself wrote a book about it: The Third Wave. When those of us aboard Bronze Age found out that we were doomed to wander space forever, we formed a totalitarian state as well. Do you know how long it took? Five minutes. That's right. Five minutes into the all-hands meeting, the fundamental values of this totalitarian society had received the support of the vast majority of the crew. So, let me tell you, when humans are lost in space, it takes only five minutes to reach totalitarianism.

Highlight (Yellow) | Page 115

ROVINSKI: Nothing special was done. They were mixed up with the vegetables and meats in the bio-recycling system and then cooked. JUDGE: Who ate this food? ROVINSKI: Everyone. Everyone onboard Bronze Age had to eat in one of the four mess halls, and there was only one source of food. JUDGE: Did they know what they were eating? ROVINSKI: Of course.

Highlight (Yellow) | Page 116

I don't have much to say except a warning. Life reached an evolutionary milestone when it climbed onto land from the ocean, but those first fish that climbed onto land ceased to be fish. Similarly, when humans truly enter space and are freed from the Earth, they cease to be human. So, to all of you I say this: When you think about heading into outer space without looking back, please reconsider. The cost you must pay is far greater than you could imagine. * * * In the end, Captain Neil Scott and six other senior officers were convicted of murder and crimes against humanity and sentenced to life imprisonment. Of the remaining 1,768 members of the crew, only 138 were declared innocent. The rest received sentences ranging from twenty to three hundred years.

Highlight (Yellow) | Page 120

Now, Cheng Xin was the only one who held legal title to a star. Though humanity still hadn't begun to reach for any star beyond the Solar System, the rapid pace of technological progress meant that stars within three hundred light-years of the Earth were no longer of mere symbolic value. DX3906, Cheng Xin's star, turned out to have planets after all. Of the two planets discovered so far, one seemed very similar to Earth based on its mass, orbit, and a spectrum analysis of its atmosphere. As a result, its value rose to stratospheric heights. To everyone's surprise, they discovered that this star already had an owner. The UN and the Solar System Fleet wanted to reclaim DX3906, but this couldn't be done legally unless the owner agreed to transfer the title. Thus Cheng Xin was awakened from her slumber after 264 years of hibernation.

Highlight (Yellow) | Page 121

A Ph.D. in astronomy named 艾 AA3 had discovered the planets around DX3906. As part of her dissertation, AA had developed a new technique that used one star as a gravitational lens through which to observe another. To Cheng Xin, AA resembled a vivacious bird fluttering around her nonstop.

Highlight (Yellow) | Page 122

"There's almost three centuries separating us. I don't expect us to understand each other right away." "Fine." AA sighed. "But you should reevaluate your conception of duty and conscience. Duty drove you to give up the planets, and conscience made you keep the star, but duty again made you give up the star's energy output. You're one of those people from the past, like my dissertation advisor, torn by conflict between two ideals. But, in our age, conscience and duty are not ideals: an excess of either is seen as a mental illness called social-pressure personality disorder.

Highlight (Yellow) | Page 123

Cheng Xin asked the question that had been puzzling her for days. “What happened to the men?” In the few days since she had been awakened, she had not seen a single man. “What do you mean? They’re everywhere.” AA pointed at the people around them. “Over there: See the man leaning against the balustrade? And there are three over there. And two walking toward us.” Cheng Xin stared. The individuals AA indicated had smooth, lovely faces; long hair that draped over their shoulders; slender, soft bodies—as if their bones were made of bananas. Their movements were graceful and gentle, and their voices, carried to her by the breeze, were sweet and tender. ... Back in her century, these people would have been considered ultra-feminine.

Highlight (Yellow) | Page 123

Understanding dawned on her after a moment. The trend had been obvious even earlier. The decade of the 1980s was probably the last time when masculinity, as traditionally defined, was considered an ideal. After that, society and fashion preferred men who displayed traditionally feminine qualities.

Highlight (Yellow) | Page 124

The strategic balance between the humans and the Trisolarans as a result of dark forest deterrence had shocked her the most. A thought popped into her mind. A world dedicated to femininity ... but what does that mean for deterrence?

Highlight (Yellow) | Page 129

Without a doubt, Luo Ji’s creation of dark forest deterrence against Trisolaris was a great achievement, but the Wallfacer Project that led to it was judged a ridiculous, childish act. Humanity, like a child entering society for the first time, had lashed out at the sinister universe with terror and confusion. Once Luo Ji had transferred control of the deterrence system to the UN and the Solar System Fleet, everyone thought the Wallfacer Project, a legendary bit of history, was over.

Highlight (Yellow) | Page 129

a new field of study was born: deterrence game theory.

Highlight (Yellow) | Page 129

The main elements of deterrence are these: the deterrer and the deteree (in dark forest deterrence, humanity and Trisolaris); the threat (broadcasting the location of Trisolaris so as to ensure the destruction of both worlds); the controller (the person or organization holding the broadcast switch); and the goal (forcing Trisolaris to abandon its invasion plan and to share technology with humanity).

Highlight (Yellow) | Page 130

people soon discovered a discouraging fact: If the authority to carry out the threat in dark forest deterrence is held by humanity as a whole, then the degree of deterrence is close to zero.

Highlight (Yellow) | Page 130

Depending on his mental state at the moment deterrence failed, Luo Ji's degree of deterrence hovered between 91.9 percent and 98.4 percent. Trisolaris would not gamble with such odds. Of course, such careful analysis wasn't possible immediately after the creation of dark forest deterrence. But humanity quickly reached this conclusion intuitively, and the UN and the Solar System Fleet handed the authority to activate the deterrence system back to Luo Ji like a hot potato. The entire process from Luo Ji turning over the authority to getting it back took a total of eighteen hours. Yet that would have been long enough for droplets to destroy the ring of nuclear bombs surrounding the Sun and deprive humanity of the ability to broadcast their locations. The Trisolarans' failure to do so was widely considered their greatest strategic blunder during the war, and humanity, covered in cold sweat, let out a held breath.

Highlight (Yellow) | Page 132

Experts believed that although science and technology usually contributed to the elimination of totalitarianism, when crises threatened the existence of civilization, science and technology could also give birth to new totalitarianism.

Highlight (Yellow) | Page 133

After the sophons unblocked the progress of human technology and the Trisolarans began transferring their knowledge to humans, human science had advanced by leaps and bounds.

Highlight (Yellow) | Page 134

The police told her that the last murder case in the world had occurred twenty-eight years ago, and this city hadn't had a murder case in almost forty years. The police were out of practice when it came to the prevention and detection of murder, and that was why Wade had almost succeeded.

Highlight (Yellow) | Page 141

They had achieved wa kei sei jaku—harmony, respect, purity, and tranquility, the four principles of the Way of Tea.

Highlight (Yellow) | Page 148

Once Gravity left the Oort Cloud behind, the ship entered true outer space. From here, the sun appeared as just another star behind the ship. It lost its reality and became like an illusion. In every direction, all one could see was a bottomless abyss, and the only objects whose existence could be ascertained by the senses were the droplets flying in formation with Gravity. The droplets flanked the ship at a distance of about five kilometers, just visible to the naked eye. Those aboard Gravity liked to gaze at the droplets with telescopes as a source of comfort in the endless emptiness.

Highlight (Yellow) and Note | Page 150

cosmos

The universe

Highlight (Yellow) | Page 154

Based on the condition of the rupture, the micrometeoroid had struck the tube at a relative velocity of thirty thousand meters per second.

Highlight (Yellow) | Page 159

Plaza #1 was located at the bow of the cylindrical ship, under a semispherical, transparent dome. Standing there was like standing in space itself. Compared to the holographic projections of the star field on the walls of spherical cabins, this place induced an even stronger sense of the “desubstantiation effect.”

Highlight (Yellow) | Page 159

in deep space, away from the Solar System, the stars were only distant points of light and the galaxy was nothing more than a luminous mist. To the senses and the mind, the world lost its materiality, and empty space dominated. A space voyager’s subconscious image of the world thus became desubstantiated. This mental model was the baseline in astronautic psychology.

Highlight (Yellow) | Page 166

At the time of the founding of deterrence, humankind already had a basic grasp of the technology for detecting gravitational waves and neutrinos, but they lacked the ability to modulate and transmit. These were the very first technologies humans demanded from Trisolaris. Compared to quantum communications, these technologies were still primitive, since both gravitational waves and neutrinos were limited by the speed of light, but they were a whole level above electromagnetic waves.

Highlight (Yellow) | Page 166

Neutrinos, in particular, interacted with almost nothing else. Theoretically, a modulated beam of neutrinos could transmit information to the other end of the universe, and the accompanying decay and distortion would not affect the decoding of the information. But while neutrinos must be focused in a particular direction, gravitational waves were omnidirectional, thus gravitational waves became the main method of establishing dark forest deterrence.

Highlight (Yellow) | Page 168

Everyone understood, however, that burying the transmitters and the control center beneath the Earth’s surface primarily guarded against a threat originating from humankind itself, but was meaningless against an attack by

Trisolaris. For droplets constructed from strong-interaction material, tens of kilometers of rock were not much different from mere liquid, barely able to impede their progress.

Highlight (Yellow) | Page 168

Trisolarans were no longer creatures of transparent thought. During the past two centuries, they had learned a great deal about strategic thinking—lies, ruses, and tricks. This was perhaps the greatest benefit they gained from studying human culture.

Highlight (Yellow) and Note | Page 169

1/10 AU of the Earth, or fifteen million kilometers. Outside this sphere, the droplets were free to roam undetected. At maximum speed, a droplet could cross fifteen million kilometers in ten minutes.

9,320,521 mi

Highlight (Yellow) | Page 170

It was said that Master Batuo, the founder of Shaolin Monastery, had meditated in front of a wall for ten years until his shadow was carved into the stone. If so, Luo Ji could have inscribed his own shadow into this wall five times.

Highlight (Yellow) | Page 172

Throughout this time, Luo Ji had remained silent, not uttering a single word. As a matter of fact, after a person ceased to speak for ten or fifteen years, he lost his powers of speech. He might still be able to understand language, but he would not be able to speak. Luo Ji certainly could no longer speak; everything he had to say, he put into his gaze against the wall. He had turned himself into a deterrence machine,

Highlight (Yellow) | Page 177

light streaked across the sky, crashing into the sea and trailing long tails of fire; these meteor strikes caused tsunamis that propelled gigantic waves to smash onto continents still laced with rivers of lava, raising clouds of vapor generated by fire and water that dimmed the Sun.... In contrast to this hellish but magnificent sight, the turbid water brewed a microscopic tale. Here, organic molecules were born from lightning flashes and cosmic rays, and they collided, fused, broke apart again—a long-lasting game played with building blocks for five hundred million years. Finally, a chain of organic molecules, trembling, split into two strands. The strands attracted other molecules around them until two identical copies of the original were made, and these split apart again and replicated themselves.... In this game of building blocks, the probability of producing such a self-replicating chain of organic molecules was so minuscule that it was as if a tornado had picked up a pile of metallic trash and deposited it as a fully-assembled Mercedes-Benz. But it happened, and so, a breathtaking history of 3.5 billion years had begun.

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But the deeper root cause could only be found in the minds of humankind. Never explicitly stated, and perhaps not even consciously understood, a gravitational wave ship was too powerful—so powerful that it terrified its creator. If something—a droplet attack or something else—forced such ships to depart for deep space, and they could never return to the Solar System due to the presence of enemy threats, they would turn into copies of Blue Space and Bronze Age, or something even more horrific. Each gravitational wave ship, with its no-longer-human crew, would also possess the power to broadcast to the universe (though limited by the half-life of the vibrating string), thus controlling the fate of humanity. A frightful instability would be permanently scattered among the stars. At its root, this fear was a fear of dark forest deterrence itself. This was characteristic of ultimate deterrence: The deterrer and the deteree shared the same terror of deterrence itself.

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“I’m leaving,” said Sophon. “Please convey to Dr. Luo Ji the deepest respect of all of Trisolaris. He was a powerful deterrer, a great warrior. Oh, and if you get the chance, also give Mr. Thomas Wade our regrets.” Cheng Xin looked up, surprised. “In our personality studies, your degree of deterrence hovered around ten percent, like a worm wriggling on the ground. Luo Ji’s degree of deterrence was always around ninety percent, like a fearsome cobra poised to strike. But Wade—” Sophon gazed at the setting sun behind the smoke, only a sliver of which now was above ground. Terror glinted from her eyes. She shook her head vigorously, as though trying to chase away a mirage in her mind. “He had no curve at all. No matter what the other environmental parameters were, his degree of deterrence stayed at one hundred percent! What a devil! If he had become the Swordholder, none of this would have been possible. This peace would have had to last. We’ve already waited sixty-two years, but we’d have had to keep on waiting, maybe for fifty more years, or even longer. And then Trisolaris would face an Earth equal to us in technology and power. We’d have to compromise.... However, we knew that humanity would choose you.”

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Only one explanation for this curious decision came to mind: It was motivated by a desire not to harm either solar system during the process of entering and dropping out of lightspeed. The safe radius appeared to be two hundred times the distance from the Sun to Neptune. This suggested that the power generated by the drives was two orders of magnitude greater than a star, which seemed unimaginable.

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he rarely talked about the crisis at all. All he said was, “Whatever people do, the gods remember.”

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He picked up a taiaha from the corner of the house and began to dance a real war haka. Cheng Xin and AA were mesmerized. Fraisse’s kind everyday demeanor disappeared, and he transformed into a threatening, awe-inspiring demon. His whole body seemed suffused with magnificent force. Every cry and foot stomp made the glass window panes quake in their frames, and the two women trembled. But it was his eyes that shocked them

the most: Murderous chill and searing rage spewed from those wide-open orbs, combining the forces of typhoons and thunder in Oceania. His powerful gaze seemed to project earth-shattering shouts: “Do not run away! I will kill you! I will eat you!”

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The haka over, Fraisse went back to his usual kind self. “For a Māori warrior, the key is to hold the enemy’s gaze. He must defeat the enemy first with his eyes, then kill him with the taiaha.” He came back and stood in front of Cheng Xin. “Child, you failed to hold the enemy’s gaze.” Then he patted her gently on the shoulder. “But, it’s not your fault. Really not your fault.”

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Though technology had continued to improve, the maximum speed of the ships never increased, as fusion propulsion had already been pushed to the limit. The overwhelming advantage the Trisolar ships held was not only their ability to reach lightspeed, but, more terrifyingly, their ability to leap into lightspeed without a prolonged process of acceleration. In order to reach even 15 percent of lightspeed, human ships had to accelerate for a year, taking into account fuel consumption rates and the need to reserve fuel for the return voyage. Compared to Trisolaran ships, Earth ships were slow as snails.

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Other than the Martian base, there was no more human presence in space in the Solar System, save for empty cities and ships drifting in orbit around the Earth, Mars, Jupiter, and through the asteroid belt. They seemed to form a silent, metallic graveyard, where humankind’s glory and dreams were buried.

Highlight (Yellow) | Page 204

Cheng Xin sat down next to Fraisse. She liked being with the old man. His transcendence in the face of a miserable reality soothed the pain of her broken heart.

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“I’m an orthopedist. You probably know that when a bone is broken, it heals stronger because a knot forms around the fracture. The body, when given an opportunity to make up for an absence, may do so excessively, and recover to the point where it has more of that quality than those who had never suffered such inadequacy.” He pointed up at the sky. “Compared to humans, the Trisolarans once lacked something. Do you think they also overcompensated? To what extent? No one knows.”

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Sophon’s words after she killed those people at the food distribution center—“The era for humanity’s degenerate freedom is over”—became a common slogan, and

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discarded dregs from the history of ideas, including fascism, crawled out of their tombs to the surface and became mainstream. The power of religions also recovered, and people gathered into different faiths and churches. Thus, theocracy, a zombie even more ancient than totalitarianism, reanimated itself. War was the inevitable result of totalitarian politics. Conflicts between nations became more frequent. At first, the conflicts were over food and water, but they soon evolved to planned contests over living space.

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But when one viewed the three-dimensional world from four-dimensional space, all concealed and hidden details were revealed simultaneously, since three-dimensional objects were laid open at every level. Take a sealed container as an example: One could see not only what was inside, but also the interiors of the objects inside. This boundless disclosure and exposure led to the unlimited details on display.

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The difficulty of describing high-dimensional spatial sense lay in the fact that for observers situated in four-dimensional space, the space they could see was empty and uniform, but there was a depth to it that could not be captured by language. This depth was not a matter of distance: It was bound up in every point in space. Guan Yifan's exclamation later became a classic quote: "A bottomless abyss exists in every inch."

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Look over there—" Chu Yan pointed in the direction they had come from. Morovich and Guan saw those tubes again, which now also lay open so that all the liquids flowing through them could be seen clearly. Inside the spherical region, the tubes were curved and distorted, and the sphere appeared like a drop of dew hanging on a spiderweb. This was different from how the region had appeared in three-dimensional space. There, the warped point didn't refract light, and so appeared completely invisible. Its presence could only be made known via the disappearance of objects that had entered four-dimensional space inside the bubble. "If you come here again, you must wear space suits. Novices can't always tell locations accurately, and finding a new warped point for the return trip might result in landing outside the ship in three-dimensional space." Chu Yan indicated that the other two should follow, and entered the dewdrop-like bubble. In an instant they were back inside the three-dimensional world, back in the corridor inside the ship, in exactly the place where, ten minutes ago, they had entered four-dimensional space. In fact, they had never left—the space they were in had just gained an extra dimension. The round opening in the bulkhead was still there, and they could still see the "broken" tubes inside.

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Morovich was feeling claustrophobic, as though he was being suffocated. "This is normal. You'll get used to it after a few times." Chu Yan laughed. "The two of you now know what real vastness means. Even if you put on space suits and go take a walk in space now, you'll feel confined."

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“Let me propose a model,” said Guan excitedly. “The entirety of our three-dimensional space is a large, thin sheet of paper, sixteen billion light-years across. Somewhere on this sheet of paper is a tiny, four-dimensional soap bubble.”

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from four-dimensional space. No matter how powerful the strong-interaction probes were, they were still three-dimensional objects. In a sense, three-dimensionality is synonymous with fragility. Viewed from four-dimensional space, it was an unrolled painting, defenseless. We approached it from the fourth dimension, and, without understanding its principles of operation, sabotaged its internal mechanisms—completely exposed—at random. “Is Trisolaris aware of the existence of the four-dimensional fragment?” “We think not.”

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Based on preliminary investigation, we think the three-dimensional projection is spherical. If that’s true, based on the data collected so far, its radius is between forty to fifty AU.” “About the size of the Solar System.” The round opening in the bulkhead next to the three of them began to move slowly and to shrink. When it was just over ten meters from them, the opening disappeared completely.

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What they were seeing wasn’t just a Ring, but an infinity of Rings all stacked together in concealment. This sensation of four-dimensionality impressed itself upon the soul, and gave the observers the experience of seeing the mountain contained in a mustard seed described in Buddhist parables.

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There is no dark forest state between spaces of different dimensions. The lower-dimensional space cannot threaten the higher-dimensional space, and the resources of the lower-dimensional space are of no use to the higher-dimensional space. But the dark forest exists everywhere between those sharing the same dimensions.

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By their calculations, a few tens of grams of four-dimensional matter could form a line stretching almost ten thousand kilometers in three dimensions. Based on the rate at which the boundary of the fragment was receding, in twenty days or so, the Ring would also enter three-dimensional space. The two ships decided to wait to observe such a wonder of the universe—

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The old model for dark forest theory had always assumed planetary systems around stars as the foundation. People simply assumed that attacks on systems whose coordinates had been exposed must come from other

planetary systems. But once the possibility of attacks from spacecraft entered the scene, the situation became far more complex. While the locations of stars were relatively well known, humans had no information at all concerning spacecraft made by other intelligences—

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Yes, she (or they?) could have succeeded in wiping them out, and had almost succeeded several times. But each time, humanity had snatched victory from the jaws of defeat through tenaciousness, cunning, and luck. After a three-century-long march, all Sophon had managed was to see her home annihilated in a sea of flames.

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“Dark forest attacks all share two qualities: one, they’re casual; two, they’re economical.” “Elaborate, please.” “These attacks are not part of some interstellar war, but a matter of conveniently eliminating possible threats. By ‘casual,’ what I mean is that the only basis for the attack is the exposure of the target’s location. There will be no reconnaissance or exploration conducted against the target beforehand. For a supercivilization, such exploration is more expensive than a blind strike. By ‘economical,’ what I mean is that the attack will employ the least expensive method: using a small, worthless projectile to trigger the destructive potential already present in the target star system.” “The energy within the stars.” Sophon nodded. “That is what we’ve seen so far.”

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“This is a gesture of respect from our world to you. My answer will be true, even if the answer could cause harm to Trisolarans. But you have only one question, and my answer must be from those three choices. Please consider it carefully before you speak.” Cheng Xin gazed at Luo Ji anxiously, but the latter didn’t pause at all. In a decisive tone, he said, “I’ve considered it. Here’s my question: If Trisolaris showed certain signs of being dangerous when observed from a distance, does there exist some sign that can be shown to the universe to indicate that a civilization is harmless and will not threaten anyone else, thus avoiding a dark forest strike? Can Earth civilization broadcast such a ‘safety notice,’ if you will, to the universe?”

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The Conversation of the Way of Tea gave the tiniest bit of hope for the expectant mass of humanity: It was possible to broadcast a safety notice to the cosmos to avoid dark forest strikes.

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Once, when humankind had been far more naïve, some scientists had believed that it was possible to detect the presence of distant civilizations by astronomical observation: for instance, the absorption spectral signatures of oxygen, carbon dioxide, and water vapor in exoplanetary atmospheres, or electromagnetic emissions. They even came up with whimsical notions like searching for signs of Dyson spheres. But we found ourselves in a universe in which every civilization endeavored to hide itself. If no signs of intelligence could be detected in a solar system from far away, it was possible that it really was desolate, but it was also possible that the civilization there had truly matured.

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But a safety notice was different: If such a notice could make the whole universe believe the Earth was harmless, then, by definition, the Earth would be harmless toward the Trisolarans. Wasn't this just what they wanted?

False proposition. They be no less inclined to revenge, even if others thought they were harmless

Highlight (Yellow) | Page 283

time. The six droplets in the Solar System had left six years ago, and there were no more blocks on the Sun's amplification function, but

Highlight (Yellow) | Page 284

Faced with the omnipresent threat of death and the lure of a different state of existence, religion once again took center stage in social life.

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Christians had to wrestle with the fact that the aliens were not in the Garden of Eden, and God never mentioned them in Genesis.

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But the Staircase Project wasn't one of them; it had been forgotten even before the Great Ravine. In the eyes of historians, the Staircase Project was a typical result of the ill-thought-out impulsiveness that marked the beginning of the Crisis Era, a hastily conducted, poorly planned adventure. In addition to the complete failure to accomplish its objectives, it left nothing of technological value.

Yet the 2nd Swordholder was it's creative mind...

Highlight (Yellow) and Note | Page 293

thousand kilometers above sea level,

621miles

Highlight (Yellow) and Note | Page 293

the terminal station, thirty thousand kilometers away,

18,641miles

Highlight (Yellow) and Note | Page 293

fifteen hundred kilometers per hour, equivalent to a supersonic jet.

932mph

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One could even walk up to the orbit of the moon in this manner, though it would be impossible to step onto the moon—by then, the relative velocity of the moon with respect to the climber would be more than three thousand kilometers per hour,

1864mph

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Cheng Xin knew that one of these housed the headquarters of the Halo Group, her space construction company. AA was working in it right now, but she couldn't tell which building it was.

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Each ring's method of construction and decorative style formed a time capsule of an age: the repressive militaristic uniformity of the Great Ravine; the optimism and romanticism of the latter half of the Crisis Era; the hedonistic freedom and indolence of the Deterrence Era. Before the fourth ring, the cabins in the rings were integrated into the structure of the rings, but starting with the fifth ring, the rings only provided construction spaces, and the buildings in the rings were planned and constructed later as additional fixtures, showing a rich variety of styles. As Cheng Xin descended through the rings, signs of this being a space station gradually faded, and the environment resembled daily life on the surface more. By the time they reached the eighth ring, the outermost ring of the station, the construction style and scenery were indistinguishable from a small city on the surface. The corridor looked like a bustling pedestrian promenade. Add to that the standard gravity of 1G, and Cheng Xin could almost forget that she was in space, thirty-four thousand kilometers above the Earth.

34k K== 21,127mi

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One side of the hall was completely open to space and the stars spinning around the station. Not too far away from them, a bright light started to glow, illuminating the whole port. Gradually, the light turned from orange to pure blue, and the spaceship that had turned on its engines lifted off the floor, accelerated, and shot into space from the open side of the port. Cheng Xin was witnessing a technological miracle that had become commonplace for others, but she couldn't figure out how it was possible to maintain atmosphere and pressure in space without the area being completely enclosed.

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Cheng Xin examined the dinghy before her. Short-range spacecraft now came in a variety of shapes, but the streamlined profile popular in the imagination of past generations was absent. This dinghy had the most common shape: a sphere. It was so regular that Cheng Xin couldn't even tell where the thruster was. The dinghy was about the size of an old medium-sized bus. It had only a serial number and no name. This common vehicle was going to carry her to the meeting with Yun Tianming.

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The meeting was to take place at the point where the Earth's and the Sun's gravities balanced each other: a Lagrangian point about 1.5 million kilometers away. The sophons would facilitate the meeting with their real-time link with the First Trisolaran Fleet. There would be both voice and video. Why conduct the meeting in space? In an age where neutrino communication was possible, being in space wasn't much more isolated than being on the surface of the Earth. Sophon had explained the request as symbolic: The meeting should occur in an isolated environment to show that it was independent of both worlds. The Lagrangian point was chosen to allow Cheng Xin's position to be relatively stable. Also, it was the long-held custom among Trisolarans to conduct meetings at points of balance between celestial bodies.

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1.5 million kilometers, or one hundredth of an astronomical unit.

932,059mi

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This was the Lagrangian point. Here, the dinghy was a satellite of the Sun, orbiting in synch with the Earth.

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The space around her was empty, and she struggled to empty her mind as well. She was preparing herself for the task of memorization: The only thing that could retain anything from the meeting was her brain. She had to turn herself into an emotionless audio and video recorder so that during the next two hours she could remember as much as possible of what she saw and heard. She imagined the corner of space she happened to be in. Here, the Sun's gravity overcame the Earth's, reaching balance, so this place held an extra measure of emptiness compared to other spots in space. She was in this emptiness of zero, a lonely, independent presence that had no connections to any other part of the cosmos.... In this way, she managed to drive her complicated emotions out of her consciousness, until she achieved the blank, transcendent state she wanted.

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This kingdom had no stories. For a kingdom, not having any stories was a good thing. The people of such a kingdom were the happiest. Stories meant twists and catastrophes.

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“This is Prince Deep Water and Princess Dewdrop. Watch your words and acts!” Guardian Shaded-Forest shouted at the officer. The officer knelt down on one knee and bowed his head. “We know. But our orders are to pursue and kill the princess.” “Princess Dewdrop is the heir to the throne by law! But Ice Sand is a traitor, guilty of regicide and patricide! How can you follow his orders?” “We know this as well, which is why we will not carry out this order. But Prince Ice Sand ascended to the throne yesterday afternoon. We ... are uncertain whose orders we should obey.” Shaded-Forest was about to say more, but Prince Deep Water stepped forward and stopped him. The prince turned to the officer. “Why don’t the princess and I return to the palace with you? We’ll confront Ice Sand there and resolve this once and for all.”

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“I saw Prince Deep Water,” said Needle-Eye, his eyes lowered. “But I couldn’t paint him.” “Did he have wings?” Ice Sand’s voice was chilly. “Even if he did, I could still capture him. I could paint each feather in his wing and make it lifelike. But, my king, the truth is more frightening than if he had sprouted wings: He does not obey the laws of perspective.” “What is perspective?” “The principles of perspective dictate that objects farther away appear smaller than those up close. I am a painter trained in Western traditions, and Western painting follows the rules of perspective. I cannot paint him.” “Are there schools of painting that do not follow the rules of perspective?” “Indeed. My king, look at those Eastern paintings.” Needle-Eye pointed to a brush-painting landscape scroll hanging on one of the walls in the hall. The scroll showed an elegant, ethereal landscape where the negative space, the emptiness, resembled water and fog. The style contrasted sharply with the colorful, solid oil paintings nearby. “You can tell that the scroll does not obey the laws of perspective. But I never studied Eastern painting. Master Ethereal refused to teach me—perhaps he had foreseen today.”

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The captain looked at the princess and let out a long sigh of relief. He turned to the ashes. “It’s too bad. The portrait was lovely, and I would have liked you to see it. But I dared not delay ... it was really, really beautiful.” “Prettier than me?” “It was you.”

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“What? But Prince Deep Water already announced that your coronation is tomorrow. He pledged to aid you with all his heart.” The princess shook her head. “My brother is more suited to be king than I. And if he hadn’t been imprisoned on Tomb Island, he ought to have inherited the throne. When he’s the king, he can stand somewhere high in the palace, and the entire kingdom can see him. But I don’t want to be a queen. I like the outside more than the palace. I don’t want to live the rest of my life in the Storyless Kingdom. I want to go where there are stories.” “That life is full of danger and hardship.” “I’m not afraid.” The princess’s eyes glowed with the spark of life in the candlelight. Long-Sail felt everything around him growing brighter again. “I’m not afraid, either. Princess, I will follow you to the end of the sea, to the end of the world.” “Then we’ll be the last two to leave the kingdom.” The princess reached out and grabbed the two floating bars of soap. “We’ll take a sailboat.” “Yes, with snow-white sails.”

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For instance, the idea of painting people into pictures was, by consensus, a rather obvious metaphor. But experts in different fields could not agree on a single interpretation. Some believed that the paintings were a reference to the digitization trend in the modern world, and thus this detail in the story suggested that humans should also be digitized as a way to avoid dark forest strikes. Scholars who held this view also noted that those who had been painted into the paintings were no longer able to harm those in the real world, and so digitizing humanity was perhaps a way to promulgate the cosmic safety notice. But another camp held that the paintings were intended to suggest special dimensions. The real world and the world of the paintings were of different dimensionalities, and when a person was painted, that person disappeared from three-dimensional space. This brought to mind the experiences of Blue Space and Gravity in the four-dimensional fragment, and so perhaps Tianming had intended to suggest that humanity could use four-dimensional space as a refuge, or broadcast the cosmic safety notice in some manner through four-dimensional space. Some scholars pointed to Prince Deep Water's violations of the rules of perspective as further evidence that the author meant four-dimensional space.

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As another example, what was the meaning of the glutton fish? Some focused on their large numbers, their habit of remaining hidden, and their fierce, aggressive tendency, and reached the conclusion that they symbolized cosmic civilization as a whole in the dark forest state. The soap that allowed the glutton fish to feel so comfortable as to forget to attack represented some unknown principles behind the cosmic safety notice. Others, however, reached the opposite conclusion: They believed that the glutton fish represented intelligent machines that must be built by humankind. These machines would be small in size and capable of self-replication. Once released into space, the machines would use the matter found in the Kuiper Belt or the Oort Cloud to self-replicate in large numbers until they formed an intelligent barrier around the Solar System. The barrier could have multiple functions, e.g., intercepting photoids headed for the Sun, or altering the appearance of the Solar System from a distance in a manner that would achieve the goal of a cosmic safety notice. This explanation, dubbed the Shoaling Interpretation, was given more attention than other competing interpretations. Compared to the other hypotheses, the Shoaling Interpretation offered a relatively clear technical framework and became one of the first interpretations to be treated as an in-depth research topic by the World Academy of Sciences. But the IDC never put too much hope in the Shoaling Interpretation—although the idea seemed technically feasible, further study revealed that it would take tens of thousands of years for the self-replicating “shoals of fish” to form a barrier around the Solar System. Moreover, the limited functionality of AI machines meant that the protective and safety notice functions of the barrier were at best impractical visions. Ultimately, the Shoaling Interpretation had to be abandoned.

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At first, everyone strove to solve the riddle for the future of humanity as a whole. But later, various political forces and interest groups began to make themselves felt: Fleet International, the UN, the various nation states, multinational corporations, religions, and so on. All of these groups tried to interpret the stories according to their own political aims and self-interest, and treated the work of interpretation as just another opportunity to disseminate propaganda about their brand of politics. The stories turned into empty baskets capable of carrying any goods.

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Viewed at the scale of the cosmos, the destruction of Trisolaris had occurred right next door, giving humans a chance to observe in detail the complete process of the extinction of a star and to gather massive amounts of data. Since the star that was destroyed was very similar to the Sun in terms of mass and position in the main sequence, humanity could potentially create a precise mathematical model of the catastrophic failure of the Sun in the event of a dark forest strike. As a matter of fact, this research had begun in earnest as soon as those on Earth had witnessed the end of Trisolaris. The direct result of research in this direction was the Bunker Project, which took the place of Yun Tianming's message as the focus of international attention.

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I. Projected timeframe from exposure of Earth's coordinates to dark forest strike against the planet: optimistic scenario, one hundred to one hundred fifty years. Average scenario: fifty to eighty years. Pessimistic scenario: ten to thirty years. Plans for the survival of the human race used seventy years as a benchmark.

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V. The Bunker Project: the four gas giants could be used as barriers to avoid the solar explosion from a dark forest strike. In the shade of the four planets, away from the sun, sufficient space habitats would be constructed to house the entirety of the human population. These space cities would be located next to the planets, but would not be their satellites. Instead, they would orbit the Sun in synchrony with the planets, staying within their shadows. The plan called for a total of fifty space cities, each of which was capable of housing about fifteen million individuals. Specifically, twenty cities would be shielded by Jupiter, twenty by Saturn, six by Uranus, and four by Neptune.

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Since the space cities would not be satellites of the gas giants, but would have to stay in close proximity of the planets, they would fall toward the planets, unless propulsion systems were installed to counteract gravity and maintain their distance from them. Initially, the plan called for the space cities to be positioned at the L2 Lagrangian points, such that the space cities' orbital periods would match their respective gas giants' without needing to expend much energy. However, it was later discovered that the L2 Lagrangian points would be too far away from the gas giants to provide sufficient protection. VII. The survival of the human race in the Solar System after a dark forest strike: After the destruction of the Sun, the space cities would rely on nuclear fusion as their energy source.

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From an energy efficiency point of view, the energy supply available to humans would actually be orders of magnitude higher than the pre-strike period because the space cities would consume fusion fuel at only one-billion-billionth the rate of the Sun.

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Since the solar explosion would destroy the terrestrial planets, the mineral resources of the Solar System would be scattered in the nebula, making them easier to collect. This would make it possible for more space cities to be constructed. The only projected resource limitation on the number of space cities was water, but there was a 160-kilometer-deep ocean covering Europa, providing a source of water greater in volume than the Earth's oceans, and capable of supplying a thousand space cities with individual populations ranging from ten to twenty million.

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Overall program for the Bunker Project: It would take twenty years to build the industrial infrastructure for extracting and exploiting resources from the gas giants, and sixty years to construct the space cities. The two stages would overlap by ten years.

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Tianming's stories had now acquired a status akin to the Bible. Without realizing it, people were no longer searching for real strategic intelligence, but reassurance that they were already on the right course. Then came the unexpected breakthrough in interpreting the stories.

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Modern baths had nothing to do with bubbles. Soap and other similar toiletries had disappeared more than a century ago. Contemporary bathing practices involved two methods: supersonic waves and cleaning agents. Cleaning agents were nanorobots invisible to the naked eye.

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As the soap dissolved in the water, it lowered the surface tension of the water behind the boat. But as the tension in the water in front of the boat remained unchanged, it pulled the boat forward.

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curvature propulsion! Excerpt from A Past Outside of Time Motion Through Bending Space Space wasn't flat, but curved.

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Supposing a spaceship could somehow iron flat the space behind it and decrease its curvature, the more curved space in front of it would pull it forward. This was the idea of curvature propulsion. Unlike folding space, curvature propulsion couldn't get a spaceship to its destination instantaneously, but it would be possible to drive it asymptotically to the speed of light.

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“It’s a centrifugal governor,” said Cheng Xin. “For steam engines.” “What’s that? Some kind of control circuit?” Bi Yunfeng shook his head. “The world wasn’t electrified back when this was invented.” Cao Bin explained. “This was a device from the eighteenth century for regulating the speed of a steam engine. It’s made of two or four lever arms equipped with spherical masses at the ends and a central spindle with a sleeve—it looks just like this umbrella, except with fewer ribs. The steam engine’s operation rotates the spindle. When it spins too fast, the metal balls lift the lever arms due to centrifugal force, which pulls up on the sleeve and reduces the aperture of the throttle valve connected to the sleeve, thereby reducing the fluid entering the cylinder and the engine’s speed. Conversely, when it spins too slowly, the lever arms fall due to the weight of the metal balls—like an umbrella closing—and the sleeve is pushed down, increasing the aperture of the throttle valve and the speed of the engine.... This was one of the earliest industrial automatic control systems.” Thus was the first level of the dual-layer metaphor in the umbrella decoded. But unlike the soap-propelled boat, the centrifugal governor didn’t seem to clearly point to anything.

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Taken together with the metaphorical meanings of the umbrella, the true meaning instantly emerged: a constant speed that did not change with the frame of reference. Clearly, it referred to the speed of light. Unexpectedly, the interpreters found yet another bearing coordinate for the metaphor of the umbrella.

Highlight (Yellow) | Page 379

Death is the only lighthouse that is always lit. No matter where you sail, ultimately, you must turn toward it. Everything fades in the world, but Death endures.”

Highlight (Yellow) | Page 381

“Sixteen point seven!” Cao Bin shouted. The fire in his eyes quickly set the other eyes around him ablaze. The third cosmic velocity of the Solar System was 16.7 kilometers per second. A spacecraft from the Earth could not leave the Solar System without exceeding this limit. It was the same with light. If the speed of light through vacuum in the Solar System were reduced to below 16.7 kilometers per second, light would no longer be able to escape the gravity of the Sun, and the Solar System would become a black hole. This was an inescapable consequence of the derivation of the Schwarzschild radius of an object, even if the object was the Solar System.

Highlight (Yellow) | Page 381

Since nothing could exceed the speed of light, if light couldn’t leave the Solar System’s event horizon, nothing else could either. The Solar System would be hermetically sealed off from the rest of the universe. And therefore completely safe—as

Highlight (Yellow) | Page 382

Later, people would call a black hole formed by lowering the speed of light a “black domain.” Compared to black holes where the speed of light was unaltered, a reduced-light-speed black hole had a much larger Schwarzschild radius. The interior was not a space-time singularity, but a fairly open region.

Highlight (Yellow) | Page 383

Within a region of space fifty AU (or 7.5 billion kilometers) in radius, a physical constant had to be altered. This was dubbed God's Engineering Project. The theoretical unknowns were immense.

Highlight (Yellow) | Page 391

"First question: Say we have a light which is off. After one minute, it blinks. Half a minute later, it blinks again. Fifteen seconds later, it blinks a third time. It keeps on going like this, blinking at intervals that are half of the immediately preceding interval. I want to know how many times it will have blinked by the two-minute mark."

Highlight (Yellow) | Page 391

"Second question: Say we have a rope whose thickness is uneven. To burn it from one end to the other takes an hour. How do you use this rope to track the passage of fifteen minutes? Remember, the thickness is uneven!"

Highlight (Yellow) and Note | Page 391

"Third question: eighty-two, fifty, twenty-six. What's the next number?" "Ten!" a girl shouted.

See nextnumber.com

Highlight (Yellow) | Page 396

"What is going on with Halo?" "We're not talking about the original ship with that name, but a new miniature stellar spaceship. It can seat twenty during planetary voyages, and five for stellar flight. The board of directors agreed to build it for you, and you can use it as a mobile office at Jupiter."

Highlight (Yellow) | Page 396

Of course, in spaceships, the difference wasn't merely a matter of volume—there were small stellar spaceships, too. Compared to planetary ships, stellar ships had more advanced propulsion systems, were equipped with ecological cycling systems, and every subsystem had three or four backups.

Highlight (Yellow) | Page 398

as observations of the Second Trisolaran Fleet revealed, spacecraft powered by curvature propulsion could achieve lightspeed almost instantaneously. A lightspeed ship could go from the Earth to Jupiter in less than an hour, and the advance warning system would be more than sufficient. Powerful and wealthy individuals who possessed lightspeed ships could thus live in comfort on the Earth and then escape at the last minute, without regard for the billions left behind. This was a prospect society simply could not tolerate.

Highlight (Yellow) | Page 400

At first blush, it seemed impossible to give advance warning for photoids, because they moved at close to the speed of light. This meant that they moved almost as fast as the radiation they generated, and reached their target almost simultaneously. In other words, the observer was outside the event's light cone. But reality was a bit more complicated. Any object with rest mass could not achieve lightspeed. Although a photoid's speed approached lightspeed, it was still slightly slower than true lightspeed.

Highlight (Yellow) | Page 401

Since it wasn't massless, it couldn't avoid the gravitational attractions of nearby celestial bodies, and its path typically ended up being slightly curved. The curvature was much greater than the curvature of light through the same gravitational field. For the photoid to strike the target, its trajectory had to take this effect into account. This meant that the path traveled by the photoid was longer than the path taken by its radiation. For these two reasons, the radiation from the photoid would reach the Solar System before the photoid itself. The twenty-four-hour estimated warning period was calculated based on the maximum distance at which photoid emissions could be observed. By the time the radiation reached the Earth, the photoid itself would still be about 180 AU away. But that was merely the ideal scenario. If the photoid were launched from a nearby spaceship, there would be almost no warning—like what had happened to Trisolaris.

Highlight (Yellow) | Page 402

Take the visible light telescope, for instance: The lenses were even bigger, and the first lens's diameter had increased from twelve hundred meters to two thousand meters, big enough for a small town to fit on it. These gigantic lenses were made from materials taken directly from the asteroid belt. The first one was a medium-sized lens five hundred meters in diameter. After that was finished, it was used to focus sunlight on asteroids so that the melted rock could be made into pure glass and then formed into additional lenses. In total, six lenses floated in a ten-kilometer-long column in space, far apart from each other. The observation station itself was located at the end of the column of lenses, and could only hold a crew of two.

Highlight (Yellow) | Page 402

After this, the largest telescope in history, had completed its shakedown tests and successfully taken its first image—a star forty-seven light-years away—Widnall, the astronomer on the crew, was as excited as if he'd just had a son. Laymen did not understand that previous telescopes could only amplify the luminosity of stars outside the Solar System, not reveal any shapes. No matter how powerful the telescopes were, the stars always showed up as tiny point sources, only incrementally brighter than images taken by lesser telescopes. But now, in the view of this ultrapowerful telescope, a star showed up as a disk for the first time. Though it was small, like a Ping-Pong ball seen from tens of meters away, and one couldn't see any details in the disk, this was still an epochal moment in the history of the ancient science of visible-light astronomy. "The cataracts have been removed from the eyes of astronomy!" said Widnall dramatically, wiping his eyes.

Highlight (Yellow) | Page 404

In the unprocessed view of the telescope, the Trisolaran system looked very ordinary, just a small patch of white against the background of space, like a feather. But after the image had been processed and magnified, it appeared as a magnificent nebula that took up the entire screen. It had been seven years since the explosion of the star, so what they were seeing now was the scene three years after the explosion. Under the influence of gravity and the exploded star's angular momentum, the nebula had turned from sharp radiating rays into a soft blur of clouds, which was then flattened by the centrifugal force of the spin into a spiral. Above the nebula, the two remaining stars could be seen. One of them showed up as a disk, while the other one, more distant, remained a point of light distinguished only by its motion against the background stars. The two stars that survived the catastrophe achieved the dream of generations on Trisolaris and formed a stable double-star system,

Highlight (Yellow) | Page 404

"From a scientific perspective, 'destroy' isn't really accurate. Nothing has disappeared. All the matter that used to be there is still there, and so is all the angular momentum. It's only the arrangement of matter that has changed, like a deck of cards being reshuffled. But life is like a straight flush: Once you shuffle, it's gone."

Highlight (Yellow) and Note | Page 405

The conclusion was inescapable: Ships using curvature propulsion left behind trails as they accelerated to lightspeed. The trails apparently did not fade with time, but expanded and altered the nature of the space around them.

Sort of like a sonic boom on reaching speed of sound...

Highlight (Yellow) | Page 406

It was still unknown how long such trails would persist in space. A guess was that the trails represented some kind of distortion in space due to curvature propulsion, and might last a long time—maybe even forever. It was reasonable to conclude that Sophon had claimed Trisolaris appeared more dangerous than the Solar System when observed from a distance because of the ten-AU-diameter curvature propulsion trail left behind within the Trisolaran system, which is what caused the dark forest strike against Trisolaris to come so quickly. The trail and the broadcast of Trisolaris's location mutually provided confirmation and made the danger value of the Trisolaran system skyrocket. During the following month, Observation Unit #1 discovered six more curvature propulsion trails in different parts of space.

Highlight (Yellow) | Page 408

Superficially, research and development of lightspeed spaceflight died for obvious reasons: to avoid advance exposure of the existence of Earth civilization by the trails generated from curvature propulsion, and to prevent increasing the Solar System's danger value in the eyes of observers elsewhere in the cosmos, either of which might have led to an earlier dark forest strike. But there were deeper reasons, too. From the Common Era to the end of the Crisis Era, humanity looked at the stars with hope. But the first few steps they took toward the stars

resulted in failure and pain. The tragic Doomsday Battle revealed the extent of humanity's fragility in the cosmos, and the internecine warfare of the Battle of Darkness had injured the human spirit in equal measure. Later events, such as the judgment of Bronze Age and the hijacking of Gravity by Blue Space, resulting in the universal broadcast, all deepened these wounds and elevated the pain to the level of philosophy.

Highlight (Yellow) | Page 409

They believed that human civilization had suffered many trials before reaching a nearly ideal democratic society, but once humanity headed for space, it would inevitably regress socially. Space was like a distorting mirror that magnified the dark side of humanity to the maximum. A line from one of the Bronze Age defendants, Sebastian Schneider, became their slogan: When humans are lost in space, it takes only five minutes to reach totalitarianism.

Highlight (Yellow) | Page 411

Over a period of two months, the asteroid had been pushed from its home in the asteroid belt to here. During its voyage, some artistic engineer who had too much time on his hands had painted it with colorful bands similar to the ones on the real Jupiter, including the Great Red Spot. Overall, however, the painted asteroid did not resemble Jupiter, but some space monster with a Cyclopean red eye.

Highlight (Yellow) | Page 413

"You know what is right, and you have the courage and sense of duty to do it. This makes you extraordinary." "But?" Cheng Xin prompted. "But, you don't have the skill or the will to complete this task. We share the same ideal. I also want to see lightspeed ships built." "What are you trying to say?" "Give it to me." "Give what to you?" "Everything you own. Your company, your wealth, your authority, your position—and if possible, your reputation and glory. I will use them all to build lightspeed ships, for your ideals, and for the grandness of the human spirit."

Highlight (Yellow) | Page 418

Ten days later, Thomas Wade became the president of the Halo Group and took over all operations. Cheng Xin and AA entered hibernation. Their consciousnesses gradually faded in the cold. It felt as though they had been drifting for a long time in a river. Exhausted, they climbed onto the shore, stopped, and watched the river continue to flow before their eyes, watched as the familiar water flowed into the distance. While they stepped briefly outside the river of time, the story of humanity went on.

Highlight (Yellow) and Note | Page 423

She had to bathe herself while she still felt weak all over. As for meals, if she hadn't asked for them through that tiny holographic display, she might never have gotten to eat. Cheng Xin wasn't annoyed by this lack of solicitousness, as she had never completely adjusted to that excessively generous era where every person's every need was taken care of. She was still a Common Era woman at heart, and she felt at home here.

China... A general lack of solicituosness

Highlight (Yellow) | Page 424

the circumsolar particle accelerator was operational, and I was awakened to research basic theory. That took fifteen years. Later, the research work turned to technical applications, and I was no longer needed, so I went back into hibernation

Highlight (Yellow) | Page 425

Cheng Xin realized that the rising “mountainside” in the distance wasn’t a mountain at all, but continued to rise until it connected with the “sky.” The world was a giant cylinder, and she was standing on the inside of it. “This is Space City Asia I, in the shadow of Jupiter,” Cao Bin said. The new world that had seemed so common a moment ago now stunned her. Cheng Xin felt that she had finally, truly awakened.

Highlight (Yellow) | Page 427

Space City Asia I was one of the earliest to be built as part of the Bunker Project. It was a regular cylinder that simulated gravity with the centrifugal force generated by spinning. With a length of thirty kilometers and a diameter of seven kilometers, its usable interior surface area was 659 square kilometers, about half the size of ancient Beijing. Once, about twenty million inhabitants had lived here, but after the completion of newer cities, the population had decreased to about nine million, so that it was no longer so crowded.... Cheng Xin saw another sun appear in the sky before her. Cao Bin explained that there were a total of three artificial suns in the space city, all of them floating along the central axis, each separated by about ten kilometers. These produced energy by nuclear fusion, and brightened and dimmed on a twenty-four-hour cycle.

Highlight (Yellow) | Page 428

Ever since awakening for the first time during the Deterrence Era, Cheng Xin had never been able to integrate into the new eras, always feeling like a stranger from another time. But she could never have imagined that she would once again feel at home more than half a century later, here behind Jupiter, more than eight hundred million kilometers from the Earth.

Highlight (Yellow) | Page 429

Cheng Xin had long since noticed that the overwhelming hyperinformation age had apparently ended. Information appeared in this world in a restrained, directed manner. Was this the result of changes in the Bunker World’s political and economic systems?

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Cheng Xin shuddered—not just from the cold, but because she realized that she was completely exposed to space! But the air around her and the air pressure were real, and she could feel cold breezes. It appeared that the technology to contain an atmosphere in a nonenclosed area had advanced even further, to the point where an atmosphere could be maintained in completely open space. Cao Bin saw her shock and said, “Oh, right now we can only maintain an atmosphere about ten meters thick above ‘ground.’”

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Cheng Xin thought of the space cities as a school of tiny fish hovering in place behind a giant rock to avoid the torrents in the river. North America I, closest to Asia I, was a pure sphere. It and the cylindrical Asia I represented the two extremes of space city design. Most of the other space cities were football-shaped, though the ratios of major to minor axes were different in each. A few other space cities took on unusual shapes: a wheel with spokes, a spindle, etc.

Highlight (Yellow) | Page 431

“That’s right. We’re very close to its orbit right now. Watch your footing and don’t be scared.” Cheng Xin tried to figure out what Cao meant. She had always thought of celestial bodies as moving slowly, almost imperceptibly—as they did in most Earth-based observations. But then she remembered that the space city was not a Jovian satellite but remained stationary relative to it. Europa, on the other hand, was a satellite that moved very fast. She remembered its speed was about fourteen kilometers per second. If the space city was very close to Europa’s orbit, then ... The white sphere expanded rapidly—so fast that it seemed unreal. Europa soon took up most of the sky, and turned from a Ping-Pong ball into a giant planet. The sensation of “up” and “down” switched in an instant, and Cheng Xin felt as if Asia I were falling toward that white world. Next, the three-thousand-kilometer-diameter moon swept overhead so that for an instant, it took up the entire sky. The space city was skimming over the icy oceans of Europa, and Cheng Xin could clearly see the crisscrossing lines in that frozen landscape, like lines in a giant palm print. The air, disturbed by the passage of Europa, whipped around her, and Cheng Xin felt an invisible force dragging her from left to right—if she weren’t wearing magnetic shoes, she was sure she’d be pulled off the ground. Whatever was nearby that hadn’t been secured to the ground flew up, and a few cables attached to spaceships also drifted into the air. A terrifying rumbling came from below her—it was the immense frame of the space city reacting to the rapidly shifting gravity field of Europa. It took only about three minutes for Europa to hurtle past Asia I, and then it was on the other side of the city and began to shrink rapidly. The eight space cities in the two front-most rows all activated their thrusters to adjust their positions after the disturbance caused by Europa. Eight fireballs lit up the sky. “How ... how close was that?” Cheng Xin asked in an unsteady voice. “The closest approach, like you experienced just now, was a hundred and fifty kilometers, basically brushing right by us. We don’t really have a choice. Jupiter has thirteen moons, and it’s impossible for the space cities to avoid them all. Europa’s orbit is inclined only slightly from the equator, and so it’s very close to these cities here. It’s the main source of water for the Jovian cities, and we’ve built a lot of industry on it. But when the dark forest strike comes, all of it will have to be sacrificed. After the solar explosion, all of the Jovian moons’ orbits will shift dramatically. Maneuvering the space cities to avoid them at that time will be a very complicated operation.”

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The first was North America I, the closest city to Asia I. The main advantage of its spherical construction was that a single artificial sun at the center was sufficient to illuminate the entire city, but the disadvantage of such a design was obvious as well: The gravity changed depending on one’s latitude. The equator had the most gravity, which decreased as you went up in latitude. The polar regions were weightless. Inhabitants in the different regions had to adjust to life under various gravity conditions.

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Next, Cao Bin took Cheng Xin to Europe IV, which sported a typical football-shaped design. Its distinguishing characteristic was the lack of a common artificial sun. Every district had its own miniature fusion sun, and the tiny suns hovered about two hundred to three hundred meters high to provide illumination. The advantage of this approach was that the weightless axis could be more efficiently utilized. The axis of Europe IV was taken up by the longest—or tallest—building among all the space cities. It was forty kilometers long and connected the north and south poles of the football. Since the interior of the building was weightless, it was mainly used as a spaceport and commercial entertainment district. Europe IV had the smallest population of all the cities, only 4.5 million. It was the wealthiest city of the Bunker World. The exquisite houses illuminated by miniature suns amazed Cheng Xin. Each house came with its own swimming pool, and a few had wide lawns. Tiny white sails dotted the serene equatorial sea, and people sat on the shore, fishing leisurely. She saw a yacht sail by slowly, and it looked as luxurious as any yacht on ancient Earth. There was a cocktail party being held aboard the yacht with live musicians.... She was astonished that such life could be transplanted into the shadow of Jupiter, eight hundred million kilometers from the Earth. * * * Pacific I, on the other hand, was the antithesis of Europe IV. This was the very first city completed by the Bunker Project, and like North America I, it was a sphere. Unlike all the other Jovian cities, it did orbit Jupiter as a satellite. Millions of construction workers had lived in Pacific I during the early years of the Bunker Project.

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Pacific I had no city thrusters, and there was no artificial sun inside. It also didn't spin, so the interior was completely weightless. After entering the city, Cheng Xin saw a fairy-tale world. It was as if a broken-down but once prosperous city had lost gravity abruptly, so that everything floated in the air. Pacific I was a city in permanent night, and each building maintained illumination with a nuclear battery.

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Since there was no "up" or "down," most of the shacks were cube-shaped, with windows (which also acted as doors) on all six sides. Some were shaped as spheres, which had the advantage of being more resilient, as the drifting buildings inevitably collided against each other. There was no notion of land ownership in Pacific I because all the buildings drifted around with no permanent location. In principle, each resident had the right to use any space in the city.

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Since the buildings were densely packed inside the city, one could navigate anywhere that way, but this method of locomotion required great skill. As Cheng Xin observed the residents flitting around the dense clusters of floating buildings, she was reminded of gibbons swinging easily from branch to branch. Cheng Xin and Cao Bin drifted close to a group of homeless men gathered around an open fire.

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Due to the weightlessness, the flames did not rise up, but formed a ball of fire floating in place. The way they drank was also special. They tossed alcohol out of bottles, forming liquid spheres in the air. Then the men,

dressed in rags and with unshaven faces, drifted along with them, capturing the spheres with their mouths and swallowing.

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Above them, the dense buildings of the city floated, and flickering lights projected onto the “plaza.” Cheng Xin saw that the hull was marked with all kinds of graffiti, stretching as far as she could see. These pictures were vibrant, wild, unrestrained, wanton and full of energy. In the shifting, uncertain light, they seemed to come alive, as though they were dreams deposited from the city above.

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This was a city of poverty and homelessness, but it also possessed its own rich life, like a weightless version of the famous Song Dynasty painting, *Along the River During the Qingming Festival*. She understood that compared to the last era, the Bunker World was not at all an ideal society. The migration to the rim of the Solar System had caused some toxic social conditions, long eliminated by progress, to reemerge. This wasn’t exactly regression, but a kind of spiraling ascent, a necessary condition for the exploration and settlement of new frontiers. * * * After they left Pacific I, Cao Bin brought Cheng Xin to see a few more space cities with unusual designs.

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“world-sense.” “What?” “The sensation of being inside a world. A space city has to have ample interior volume and wide-open views so that the residents can feel they are living inside a world. Although the usable interior surface area isn’t too different from a hollow-shell design, in a wheel design, people always know that they are living inside a narrow tube or a series of such tubes.”

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The most important idea was city docking. In those two clusters at the edge of the Solar System, every city was equipped with one or more standardized docks so that cities could be interconnected. Docking multiplied the space available for inhabitants and created even better world-sense, greatly encouraging economic development. In addition, after docking, the atmospheres and ecological systems of the various cities merged, and that helped to stabilize their operation and maintenance. Currently, most cities docked along their axis of spin. This way, after docking, the cities could continue to spin as before without changing the distribution of gravity.

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People hoped that, one day, all the cities in each cluster could be combined into one, so that humanity would live in four complete worlds. In total, behind Jupiter, Saturn, Uranus, and Neptune, there were sixty-four large space cities and nearly a hundred medium and small cities, plus numerous space stations. Nine hundred million people lived in the Bunker World. This was almost the entirety of the human race. Even before the arrival of the dark forest strike, Earth civilization had battened down the hatches. Every space city was politically equivalent to a state. The four city clusters together formed the Solar System Federation, and the original

Highlight (Yellow) | Page 438

The Earth was now barely inhabited. Only about five million people remained there. These were individuals who did not wish to leave their home and who had no fear of the prospect of Death at any moment. Many brave men and women living in the Bunker World also traveled to Earth as tourists, though each journey meant gambling with their lives. As time passed, the anticipated dark forest strike loomed larger, and people gradually adapted to life in the Bunker World. Their yearning for their homeland lessened as they busied themselves in their new homes,

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The public no longer cared much about news from the home world, and were only vaguely aware that Nature was enjoying a resurgence. Forests and grasslands covered every continent, and those who stayed behind had to carry guns to defend against wild beasts when they went out, but it was rumored that they lived like kings, each with a vast estate and personal forests and lakes. The entire Earth was now only a single city in the Solar System Federation.

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“The light is caused by space dust falling into the black hole,” Cao Bin said, pointing in the direction of the light source. He was trying to relieve some of Cheng Xin’s terror. “There’s a black hole over there?” “Yes. It’s about ... no more than five kilometers from here. A microscopic black hole with a Schwarzschild radius of twenty nanometers and a mass equivalent to Leda, the Jovian moon.” In this phosphorescent blue glow, Cao Bin told Cheng Xin the story of Lightspeed II and 高 Way.⁷

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The speed of light through vacuum was one of the fundamental constants of the universe. Altering it was equivalent to altering the laws of physics. Thus, lowering the speed of light required breakthroughs in fundamental physics—and considerable serendipity. After sixty years, the only substantive result of basic research was the creation of the circumsolar particle accelerator. This, in turn, led to the success of the largest project under the Black Domain Plan: the Black Hole Project.

Highlight (Yellow) | Page 442

The circumsolar particle accelerator was capable of creating microscopic black holes. The head of the Black Hole Project was 高 Way. Cao Bin had worked with him for a few years. He could not hide his complex feelings about the man as he described him to Cheng Xin. “The man suffered from severe autism—no, I’m not talking about some kind of lonely genius choosing to isolate himself, but a real mental condition. He was extremely closed off and had trouble communicating with anyone, and he had never even touched a woman. His extraordinary professional success would only be possible in this age, but despite his accomplishments, most of his supervisors and colleagues thought of him as merely a high-powered intelligence battery. He was tortured by his illness and tried to change himself, and in this, he was different from other geniuses. “Starting from, oh, I think the eighth year of the Broadcast Era, he dedicated himself to the theoretical study of lowering the speed of light.

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“After the microscopic black hole was injected into Leda, it began to absorb mass and rapidly grow. At the same time, the intense radiation generated by material falling into the black hole melted the surrounding rock. Soon, the eight-kilometer-radius Leda melted entirely, and the potato-shaped rock turned into a red-glowing ball of lava. The lava ball shrank slowly, but glowed brighter and brighter, until it finally disappeared with a blinding flash. Observation showed that other than a small amount of material ejected by the radiation, most of the mass of Leda had been absorbed by the black hole. The black hole remained stable, and its Schwarzschild radius, or event horizon radius, had grown from the size of a fundamental particle to twenty-one nanometers. “They constructed a space city around the black hole—that’s Lightspeed II. The black hole was suspended in the middle of the space city, which was empty, didn’t spin, and whose interior was a vacuum connected to space. It was, in essence, a giant container for the black hole. Personnel and equipment could be brought into the city to study the black hole.

Highlight (Yellow) and Note | Page 445

“What happened after 高 Way was sucked in was almost too strange to describe. Scientists observed the black hole via remote-controlled microscopes, and discovered that at the black hole’s event horizon—that’s the surface of that tiny sphere with a diameter of twenty-one nanometers—there was the figure of a person. It was 高 Way passing through the event horizon. “Under general relativity, a distant observer would see a clock near the event horizon slow down, and the process of 高 Way falling toward the event horizon would also slow down and stretch into infinity. “But within 高 Way’s own frame of reference, he had already passed through the event horizon. “Even more oddly, the figure’s proportions were normal. Perhaps it was because the black hole was so small, but tidal forces did not seem to be at work. He had been compressed into the nanometer range, but space there was also extremely curved. More than one physicist believed that the body structure of 高 Way wasn’t harmed at the event horizon. In other words, he’s probably still alive at this moment. “And thus the life insurance company refused to pay out, although 高 Way had passed through the event horizon in his frame of reference, and should now be dead. But the insurance contract was made within the frame of reference of our world, and from this perspective, it is impossible to prove that 高 Way is dead. It’s not even possible to begin the settlement process. Insurance claims settlement can only occur after the conclusion of an accident, but as 高 Way is still falling toward the black hole, the accident isn’t over, and will never be over.

Per the general relativity model at the event horizon...

Highlight (Yellow) | Page 446

She even wrote her love in big letters and posted it against the protective net, hoping that the falling 高 Way could see it. But based on his own frame of reference, he had already passed through the event horizon into the singularity. “It’s a complicated matter.” Cheng Xin stared at the blue phosphorescence far away in the darkness. She now knew that there was a man there, a man who was falling forever, at the event horizon where time stopped. Such a man was still alive when viewed from this world, but had already died in his own world. “So many strange fates, and so many unimaginable lives. “

Highlight (Yellow) | Page 447

The stellar-class ships were immense, but next to the space city they appeared as tiny skiffs around a giant ocean liner. The Federation Fleet had sent the bulk of their ships to enforce this blockade of Halo City.

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The company's growth was the result of the new management team he put in place. He personally did not participate in the running of the company and had little interest in it, but much of the profit generated by the company was taken by him and reinvested in the development of lightspeed spaceflight. As soon as the Bunker Project began, the Halo Group constructed Halo City as a research center. The Sun-Jupiter L2 Lagrangian point was chosen as the ideal space to set up Halo City in order to eliminate the need for city thrusters and the consumption of resources for position maintenance.

Highlight (Yellow) | Page 449

While Halo City was being constructed, Wade also began the construction of the circumsolar particle accelerator, a project that was dubbed "The Great Wall of the Solar System" because it enclosed the Sun in a ring.

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For half a century, the relationship between the company and the government was basically cordial. Since lightspeed ships and the Black Domain Plan called for much of the same basic research, the Halo Group and the World Academy of Sciences maintained a good collaborative working relationship. For instance, the Academy's Black Hole Project used the Halo Group's circumsolar particle accelerator to produce its black hole specimen. However, six years ago, the Halo Group had suddenly announced its plan to develop curvature propulsion ships. Such open defiance caused an uproar in the international community. Thereafter, conflict between the Halo Group and the Federation Government never ceased.

Highlight (Yellow) | Page 450

The danger of lightspeed ships lay not only in the trails, but also in upsetting the new social stability in the Bunker World, a prospect that could not be tolerated.

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A ring-shaped structure appeared in space ahead, and Cao Bin ordered the dinghy to approach it and decelerate. The smooth metallic surface of the ring reflected the stars as long streaks and distorted the image of the dinghy, bringing to mind the Ring that Blue Space and Gravity had encountered in four-dimensional space. The dinghy stopped and hovered next to the ring. Cheng Xin estimated that the ring's diameter was about two hundred meters across, and the band about fifty meters thick. "You're looking at the circumsolar particle accelerator," Cao Bin said, his tone awed. "It's ... rather small." "Oh, sorry; I wasn't clear. This is but one of the coils in the particle accelerator. There are thirty-two hundred coils like this, each about one point five million kilometers

apart, forming a large circle around the Sun in the vicinity of Jupiter's orbit. Particles pass through the center of these coils, where they're accelerated by the force field generated by the coil toward the next coil, where they're accelerated again.... A particle might travel around the Sun multiple times during the process."

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Cheng Xin finally realized that while an enclosed tubular ring was necessary for terrestrial particle accelerators to maintain vacuum, it was not necessary in the vacuum of space. The particles being accelerated could simply fly through space, being accelerated by one coil after another. Cheng Xin couldn't help turning to look past the coil for the next one. "The next coil is one point five million kilometers away, four or five times the distance from the Earth to the moon. You can't see it,"

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"This is a supercollider capable of accelerating a particle to the energy level of the big bang. Ships are not allowed anywhere near the orbit of the accelerator. A few years ago, a lost freighter drifted into the orbit by mistake and was hit by a beam of accelerated particles. The ultrahigh-energy particles struck the ship and produced high-energy secondary showers that vaporized the ship and its cargo of millions of tons of mineral ore in an instant."

Highlight (Yellow) | Page 452

Halo City was straight ahead. This space city was built along the rarely seen wheel plan. The structure provided strength but had little interior volume, lacking "world-sense." It was said that the inhabitants of Halo City did not need world-sense, because for them, the world was the entire universe.

Highlight (Yellow) | Page 452

Since the simulated gravity in the ring pulled toward the outer rim, the city was built along that surface. A holographic image of blue sky with white clouds was projected onto the inner rim, which made up some for the lack of world-sense. A flock of twittering birds flitted overhead, and Cheng Xin noticed that they were not holograms, but real. Here, Cheng Xin felt a sense of comfort that she had not experienced in the other space cities. There were plenty of trees and lawns everywhere.

Highlight (Yellow) | Page 455

"What did you see?" Wade asked. "You spent half a century to move a three-millimeter section of hair two centimeters," said Cheng Xin. "It was curvature propulsion," Wade said. "If we used the same technique to continue to accelerate the hair, it would be moving at lightspeed in about ten meters," Bi Yunfeng said. "Of course we can't achieve this now, and we dare not try it here. If we did, this bit of hair, moving at lightspeed, would destroy Halo City." Cheng Xin pondered the strand of hair that had been moved two centimeters by curving space. "You are saying that you've invented gunpowder and managed to make a firecracker, but the ultimate goal is to make a space rocket. A thousand years may separate those two achievements." "Your analogy is flawed," Bi Yunfeng said. "We have invented the equation relating energy to mass, and we've discovered the

principle of radioactivity. The ultimate goal is to make the atom bomb. Only a few decades divide those two achievements.” “In fifty years, we should be able to construct curvature propulsion spaceships capable of lightspeed flight. This will require massive amounts of technical testing and development work. We have to lay our cards on the table now so that the government can back off and give us the environment necessary to carry out these tasks.”

Highlight (Yellow) | Page 456

“You can trust your eyes: ancient weapons constructed of modern materials. The bullets do not rely on gunpowder as the propellant and have much better range and precision compared to genuine ancient weapons. In space, these rifles can hit a ship from two thousand kilometers away, but fundamentally, they’re primitive weapons. You must think this ridiculous, and I would, too, except for one thing.” He returned the rifle to the soldier and pulled one of the cartridges from his bandolier. “As I said, the cartridges are basically of ancient design, but the bullets are new. So new, in fact, that they might as well come from the future. The bullet is a superconducting container and the interior is a pure vacuum. A magnetic field suspends a small ball in the middle to prevent it from contacting the bullet’s body. The ball is made of antimatter.”

Highlight (Yellow) | Page 457

The primitive-seeming cartridge held in Wade’s hand now caused Cheng Xin to suffer chills. She now worried about the reliability of the containment magnetic field within that superconducting bullet: a single malfunction would be enough to cause the complete destruction of Halo City in a brilliant flash. She looked at the golden bandoliers hanging over the chests of every soldier: These were the chains of the god of Death. A single bandolier possessed enough power to destroy the entire Bunker World.

Highlight (Yellow) | Page 457

A single soldier with a gun is a fighting unit capable of threatening an entire warship.

Highlight (Yellow) | Page 458

Also, we have agents in other space cities with handguns. If war were to erupt, any of the guerrilla fighters hidden in the other space cities could casually shoot one of the antimatter bullets at the ground and the explosion of matter-antimatter annihilation would instantaneously tear apart the thin shell of the city and incinerate everything within. Next, the spinning space city would break into fragments and millions would die. Space cities were as fragile as eggs. Wade had not explicitly said that he would attack the space cities, but he also hadn’t said he wouldn’t. She again saw Wade aiming a gun at her 133 years ago—an image which had been branded into her heart. She didn’t know how cold a man would have to be to make such a decision, but the core of this man was the utter madness and coldness brought about by extreme rationality. She seemed to see again the young Wade from three centuries ago, screaming like a crazy beast: “Advance, advance without regard for consequences!”

Highlight (Yellow) | Page 458

Another soldier spoke up. “We are not fighting for you, for Mr. Wade, or this city.” He pointed upwards, and fire seemed to light up his eyes. “Do you know what they’re trying to take away from here? Not the city or lightspeed ships, but the entire universe outside the Solar System! There are billions and billions of new worlds out there, but they won’t let us go; they want to lock us and our descendants in this prison, a prison fifty astronomical units in radius called the Solar System. We are fighting for freedom, for a chance to live as free men in the universe. Our cause is the same as every ancient struggle for freedom. We will fight to the very last. I speak for everyone in the self-defense force.”

Highlight (Yellow) | Page 462

AA said, “Why don’t we go to the world after the strike, the world after the Sun is gone? Only then will we find a life of serenity.” Cheng Xin also wanted to go to that time, but not for a life of serenity. She had stopped a catastrophic war and she was becoming the target of the worship of millions. She could no longer live in this era. She wanted to see Earth civilization survive the dark forest strike and prosper after—it was the only hope that could comfort her heart. She imagined life in that post-strike nebula. There, she would find true tranquility, maybe even happiness.

Highlight (Yellow) | Page 470

It was as the Elder said: In the cosmos, no matter how fast you are, someone will be faster; no matter how slow you are, someone will be slower.

Highlight (Yellow) | Page 470

If Singer were patient, all sincere coordinates would eventually be cleansed by other, unknown low-entropy entities. But this was not a good thing for either the home world or the seed. Since Singer had received the set of coordinates and even glanced at the world pointed to by the coordinates, Singer had a connection to that world. It would be naïve to think of this connection as unidirectional. Recall the great law of reversible discovery: If you could see a low-entropy world, then that low-entropy world could also see you—it was only a matter of time. Thus, waiting for others to complete cleansing was dangerous.

Highlight (Yellow) | Page 471

First, the other location had broadcast a message via primitive membrane. The low-entropy entities of that world clumsily plucked their star—Singer decided to call them the Star-Pluckers—like ancient bards of the home world plucking the strings of the rough country zither, to send out the message. It was this message that contained the self-decoding system. Although the self-decoding system was primitive and clumsy, it was sufficient to allow Singer to see that a subsequent message sent out by the dead three-star world followed the same encoding scheme—apparently an answer to the first message sent by the Star-Pluckers! This was already nearly inconceivable, but after that, the Star-Pluckers responded again! Interesting. Very interesting! Singer had indeed heard of low-entropy worlds that possessed neither the hiding gene nor the hiding instinct, but this was the first time he had seen one.

Highlight (Yellow) | Page 472

Hide yourself well; cleanse well.

Highlight (Yellow) | Page 474

As he continued to sing, Singer picked up the dual-vector foil with a force field feeler and carelessly tossed it at the Star-Pluckers. Bunker Era, Year 67 Halo Cheng Xin awakened to find herself in weightlessness.

Highlight (Yellow) | Page 482

白 Ice's original name was Bai Aisi.⁹ After awakening, he wanted to appear more integrated and not so behind the times, and chose a more common modern name that mixed English and Chinese elements.

Highlight (Yellow) | Page 483

this is one of them: a piece of the cooling system for a nuclear fusion reactor aboard a stellar-class warship from the end of the Crisis Era." "It's from the Doomsday Battle?" asked an awed 白 Ice. "Yes.

Highlight (Yellow) | Page 486

clearly, the sheet hadn't been broken, but passed through the hand unharmed. Vasilenko pulled his hand back, and the slip hung still as before—or, more accurately, continued to move toward the Solar System at the rate of two hundred kilometers per second. 白 Ice also tried to touch the slip, then pulled his hand back. "It's like a projection from another universe that has nothing to do with ours." Vasilenko had more practical concerns. "If nothing can affect it, then we have no way to bring it to the ship for further analysis." 白 Ice laughed. "That's a simple problem to solve. Have you forgotten the story told by Francis Bacon? 'If the mountain will not come to Muhammad, then Muhammad must go to the mountain.' " And so, Revelation slowly sailed toward the paper slip, made contact, and then allowed it to enter the ship. Even more slowly, it adjusted its position until the slip hung in the middle of the laboratory cabin. The only way to move the slip during study was to move the ship itself.

Highlight (Yellow) | Page 489

"Forget about the uncertainty principle for a minute and suppose everything is determinable. If you know the initial conditions, you can calculate and derive the conditions at any later point in time. Suppose an extraterrestrial scientist were given all data about the Earth several billion years ago. Do you think it could predict the existence of this desert solely through calculation?" Bai Aisi pondered this. "No. This desert wasn't the result of the Earth's natural evolution, but the result of man-made forces. The behavior of civilizations can't be grasped through the laws of physics." "Very good. Then why do we and our colleagues all want to try to explain the conditions of today's cosmos, and to predict its future, solely through deductions based on the laws of physics?"

Highlight (Yellow) | Page 494

“There is acceleration,” said 白 Ice weakly. “But the space in this region is flowing in the opposite direction, thus canceling out our motion.” “The space is flowing? Where to?” “There, of course.”

Highlight (Yellow) | Page 494

Do you remember what Blue Space and Gravity saw?” No one answered, but they all remembered: the four-dimensional space falling into three dimensions, like a waterfall off a cliff. “Just as four-dimensional space collapses into three dimensions, three-dimensional space can collapse into two dimensions, with one dimension folding and curling into the quantum realm. The area of that slice of two-dimensional space—it only has area—will rapidly expand, causing more space to collapse.... We’re now in space that is falling toward two dimensions, and ultimately, the entire Solar System will follow. In other words, the Solar System will turn into a painting with no thickness.” “Can we escape it?” “Escaping this is like rowing a boat above a waterfall. Unless we exceed a certain escape velocity, we’ll tumble over the cliff.

Highlight (Yellow) | Page 495

The entire Solar System is within the zone of collapse, and anyone trying to escape must reach escape velocity.” “What is the escape velocity?” “I’ve computed it four separate times. Pretty sure I got it right.” “What is it?!” Everyone aboard Revelation and Alaska held their breaths and listened to this final calculation as representatives of humanity. 白 Ice calmly announced his judgment. “Lightspeed.”

Highlight (Yellow) | Page 496

Revelation slid faster toward the plane, toward that abyss whose thickness was zero. “Everyone, don’t be sad. No one will be able to escape from the Solar System, not even a bacteria or virus. All of us will become a part of this grand picture.” 白 Ice now looked calm and stoic. “Stop accelerating,” said Vasilenko. “What difference does a few minutes make? Let’s at least breathe easier at the end.” Revelation’s engine shut off.

Highlight (Yellow) | Page 496

“You know what I’m thinking of? Needle-Eye’s pictures from Yun Tianming’s fairy tales,” 白 Ice said.

Highlight (Yellow) | Page 497

Weakness and ignorance are not barriers to survival, but arrogance is.

Highlight (Yellow) | Page 500

Halo’s other distinguishing characteristic was the ability to land directly on a medium-sized solid planet. This was a rare design choice among stellar ships—similar ships typically used shuttles to carry landing parties onto planets. Directly descending into a planet’s deep gravity well required the ship to have a very strong hull, which

greatly increased the cost. Moreover, the need for atmospheric flight required a streamlined profile, which was also very rare among stellar ships.

Highlight (Yellow) | Page 501

Halo stopped accelerating. Behind it, Jupiter became a small yellow dot. The next few days of the voyage were spent in the uninterrupted slumber produced by the sleep-aid machine. In this lonely voyage through the night before the end, just the unstoppable mad imaginings were enough to make anyone fall apart. * * * Halo's AI awakened AA and Cheng Xin from their dreamless sleep as the ship reached Pluto.

Highlight (Yellow) | Page 507

they told me that, according to the most advanced theories and techniques in every field, based on extensive theoretical research and experimentation, through analysis and comparison of multiple proposals, they did find a way to preserve information for about one hundred million years. And they emphasized that this was the only method known to be practicable. Which is—" Luo Ji lifted the cane over his head, and as his white hair and beard danced in the air, he resembled Moses parting the Red Sea. Solemnly, he intoned, "—carving words into stone." AA giggled.

Highlight (Yellow) | Page 509

What was there to say? Civilization was like a mad dash that lasted five thousand years. Progress begot more progress; countless miracles gave birth to more miracles; humankind seemed to possess the power of gods; but in the end, the real power was wielded by time. Leaving behind a mark was tougher than creating a world. At the end of civilization, all they could do was the same thing they had done in the distant past, when humanity was but a babe: Carving words into stone.

Highlight (Yellow) | Page 510

As Luo Ji had said, this wasn't a museum. A museum was built for visitors; a tombstone was built for the builders.

Highlight (Yellow) | Page 518

The drawing reflected every detail of the city, down to every screw, every fiber, every mite, and even every bacterium. The precision of the drawing was at the level of the individual atom. Every atom in the original three-dimensional space was projected onto its corresponding place in two-dimensional space according to ironclad laws. The basic principles governing this drawing were that there could be no overlap and no hidden parts, and every single detail had to be laid out on the plane. Here, complexity was a substitute for grandeur. The drawing wasn't easy to interpret—it was possible to see the overall plan of the city and recognize some big structures, such as the giant trees, which still looked like trees even in two dimensions. But buildings looked very different after being flattened: it was almost impossible to deduce the original three-dimensional structure from the two-dimensional drawing by imagination alone. However, it was certain that image-processing software equipped with the right mathematical model would be able to.

Highlight (Yellow) | Page 520

But the flattened Earth did show her own wonders. At her outermost rim, a white ring gradually appeared. At first it was barely visible, but soon it stood out sharply against the black backdrop of space. The white ring was pure, flawless, but seemed uneven in its makeup, like it was formed from countless small white grains. “That’s our ocean!” Cheng Xin said. “The water froze in two-dimensional space,” said AA. “It’s cold there.” “Oh—” Luo Ji wanted to stroke his beard, but the visor got in the way of his hand.

Highlight (Yellow) | Page 521

But what really shocked them was the frozen two-dimensional ocean—the white ring around the rim of the Earth. They could clearly make out the grains forming the ring: snowflakes!

Highlight (Yellow) and Note | Page 521

“How big are the snowflakes?” AA asked. “Most have diameters between four thousand and five thousand kilometers.” The ship’s AI, incapable of wonder, continued to speak in a serene tone. “Bigger than the moon!” Cheng Xin said.

3107mi

Highlight (Yellow) | Page 522

“How are they formed?” AA asked. “I don’t know,” the AI said. “I can’t find any information about the crystallization of water at astronomical scales.”

Highlight (Yellow) | Page 523

But it was impossible to escape from the collapsing zone. Halo’s AI was trying to say this: The three-dimensional space of the Solar System was like a large carpet that was being pulled by invisible hands into a two-dimensional abyss. These ships were nothing more than worms on the carpet inching along—they couldn’t extend their already limited allotment of time by much. “Go ahead by yourselves,” Luo Ji said. “Just take a few more objects. I want to wait here. I don’t want to miss it.” Cheng Xin and AA understood what he meant by “it,” but they had no desire to witness the scene.

Highlight (Yellow) and Note | Page 524

As soon as the Sun began to two-dimensionalize, a circle expanded on the plane. Soon, the planar Sun’s diameter exceeded the diameter of the remaining part of the Sun. This process took only thirty seconds. Based on the mean solar radius of seven hundred thousand kilometers, the rim of the two-dimensional Sun grew at the rate of twenty thousand kilometers per second. The planar Sun continued to grow, forming a sea of fire on the plane, and the three-dimensional Sun sank slowly into this blood-red sea of fire.

12427mps

Highlight (Yellow) | Page 524

While the three-dimensional Sun was setting, the two-dimensional Sun was rising. A flat star could still radiate its light inside the plane, so the two-dimensional Solar System received its first sunlight. The sides of the four two-dimensional planets facing the sun—Neptune, Saturn, the Earth, and Mercury—all took on a golden glow, though the light only fell along a one-dimensional curved edge. The giant snowflakes that surrounded the Earth melted and turned into white vapor, which was blown by two-dimensional solar wind into two-dimensional space. Some of the vapor soaked up the golden sunlight and appeared as if the Earth had hair that drifted with the wind. An hour later, the Sun had completely collapsed into two dimensions. From Pluto, the Sun appeared as a giant oval.

Highlight (Yellow) | Page 527

Cheng Xin did feel an impulse to take off her gloves and catch the snow with her bare hands. She wanted to feel the slight chill, and watch the crystalline snowflakes melt with her own body heat.... but of course she had enough presence of mind to not indulge the impulse. The nitrogen-ammonia snowflakes were at a temperature of minus-210-degrees Celsius. If she really took off her gloves, her hand would turn as fragile and hard as glass and the feeling of being on Earth would disappear instantaneously. "There's no more home," Luo Ji said, shaking his head and leaning against his cane. "Home is now just a picture." The nitrogen-ammonia snow didn't last long. The snowflakes thinned out and the purple haze from the nitrogen-ammonia atmosphere faded.

Highlight (Yellow) | Page 528

Luo Ji lifted his cane in farewell and turned to walk toward the monolith. With the light gravity, he almost floated over the snow on the ground and had to use the cane to slow himself. Cheng Xin and AA watched until the aged figure of this Wallfacer, Swordholder, and humanity's final grave keeper disappeared behind the door of the monolith.

Highlight (Yellow) | Page 529

"Oh, heavens! Starry sky!" AA cried out. Cheng Xin knew that she was referring to Van Gogh's painting. True, the universe really did look like the painting. The painting in her memory was almost a perfect copy of the two-dimensional Solar System before her eyes. Giant planets filled space, the areas of the planets seeming to exceed even the gaps between them. But the immensity of the planets did not give them any sense of substantiality. Rather, they looked like whirlpools in space-time. In the universe, every part of space flowed, churned, trembled between madness and horror like fiery flames that emitted only frost. The Sun and the planets and all substance and existence seemed to be only hallucinations produced by the turbulence of space-time. Cheng Xin now recalled the strange feeling she had experienced each time she had looked at Van Gogh's painting. Everything else in the painting—the trees that seemed to be on fire, and the village and mountains at night—showed perspective and depth, but the starry sky above had no three-dimensionality at all, like a painting hanging in space. Because the starry night was two-dimensional. How could Van Gogh have painted such a thing in 1889? Did he, having suffered a second breakdown, truly leap across five centuries and see the sight before them using only his spirit and delirious consciousness? Or, maybe it was the opposite: He had seen the future, and the sight of this Last Judgment had caused his breakdown and eventual suicide.

Highlight (Yellow) | Page 529

Cheng Xin and AA looked at each other. “Go where?” AA asked. “Anywhere you like. You can go to any place in the Milky Way. In your lifetimes, you could probably get to the Andromeda Galaxy. Halo is capable of lightspeed flight. It is equipped with the world’s only curvature propulsion drive.” Utter shock. AA and Cheng Xin couldn’t speak. “I was a part of the group of scientists who worked on curvature propulsion in secret,” said Luo Ji. “After Wade died, those who had worked at Halo City didn’t give up. After those who had been imprisoned were released, they built another secret research base, and your Halo Group was revived and developed enough to keep it going. Do you know where the base was? Mercury, another place in the Solar System where few set foot. Four centuries ago, another Wallfacer, Manuel Rey Diaz, used giant hydrogen bombs to blast a crater there. The base was built in that crater, and its construction took over thirty years. The whole base was covered with a dome. They claimed that it was a research institute to study solar activity.”

Highlight (Yellow) | Page 530

“Did they repeal the laws proscribing lightspeed ships?” Cheng Xin asked. “No, not at all. The government collaborated with us because...” Luo Ji tapped his cane against the ground and hesitated. “Let’s not get into that for now. A few years ago, we completed three curvature engines and conducted three unmanned tests. Engine Number One entered lightspeed about one hundred and fifty astronomical units from the Sun, and returned here after flying at lightspeed for a while. For the engine itself, the experiment lasted only ten minutes or so, but for us, it was three years before the engine returned. The second test involved Engines Number Two and Number Three simultaneously. Right now, both of them are outside the Oort Cloud, and should return to the Solar System in six years. “Engine Number One, which has already been tested, is installed in Halo.”

Highlight (Yellow) | Page 531

The three-dimensional space they were in accelerated toward the two-dimensional plane. The two-dimensional Sun, which had now completely extinguished and appeared as a vast, dark red, dead sea, took up most of the view from Halo. Cheng Xin and AA noticed that the plane was not completely flat, but undulating! A long wave slowly rolled across the plane. It was a similar wave in three-dimensional space that had allowed Blue Space and Gravity to find warp points to enter four-dimensional space. Even in places where there were no two-dimensional objects in the plane, the rippling wave was apparent. The waves were a visualization of two-dimensional space in three dimensions that occurred only when the two-dimensional space was large enough. On Halo itself, the space-time distortion produced by the accelerated fall had started to become apparent as space was stretched in the direction of the fall. Cheng Xin noticed that the circular portholes now appeared as ovals, and the slender AA now looked short and squat. But Cheng Xin and AA felt no discomfort, and the ship’s systems were operating normally.

Highlight (Yellow) | Page 532

“You don’t need any kind of preparation. This propulsion method relies on manipulating space, so there’s no hypergravity. Curvature propulsion drive online. System is operating within normal parameters. Local space curvature: twenty-three point eight. Forward curvature ratio: three point forty-one to one. Halo will enter lightspeed in sixty-four minutes, eighteen seconds.” For Cheng Xin and AA, the AI’s announcement was like a

Full Stop order, because everything suddenly quieted down. They understood that the silence was due to the nuclear fusion engine being shut off, but the humming produced by the fusion reactor and the thrusters disappeared without being replaced by any other noise. It was hard to believe that some other engine had been started. But signs of curvature propulsion did appear. The distortion in space gradually disappeared: The portholes returned to being circles, and AA looked slender again. Looking through the portholes, they could still see other escaping ships passing by Halo, but they now passed far more slowly.

Highlight (Yellow) | Page 533

Another scream—this one from AA inside the ship. “How can there be two Plutos?” Cheng Xin turned to the information window AA was looking at. The window showed a view of Pluto taken by the ship’s monitoring system. Although Pluto was some distance away, it was clear that both Pluto and Charon had been duplicated, and the twins were lined up side by side. Cheng Xin noticed that some of the flattened objects in the two-dimensional space had also been duplicated. The effect was like selecting a portion of a picture using image-processing software, cloning it, and then moving the clone a bit to the side. “That’s due to the fact that light slows down inside the trail left by Halo,” Luo Ji said. His image was growing distorted, but his voice still came through clearly. “Pluto is still moving. One of the Plutos you are seeing is the result of slow light. Once Pluto has moved outside of Halo’s trail, light traveling at standard speed provides you with a second image. That’s why you’re seeing double.” “The light slows down?” Cheng Xin sensed a great secret was being revealed.

Highlight (Yellow) | Page 533

Luo Ji continued, “I understand that you figured out curvature propulsion from a small boat propelled by soap. Let me ask you: After the ship reached the other side of the bathtub, did you pull it back and try again?” They hadn’t. Due to the fear of sophons, Cheng Xin had tossed the paper boat aside. But it was easy to figure out what would have happened. “The ship would not move, or at least it would only move slowly,” Cheng Xin said. “After the first trip, the surface tension of the water in the tub had already been reduced.” “That’s right. It’s the same principle with lightspeed ships. The very structure of space itself is changed by the trail of a curvature-propelled ship. If a second curvature-propelled ship were placed inside the trail of the first, it would hardly move. Within the trails of lightspeed ships, one must use a more powerful curvature propulsion drive. It would still be possible to use curvature propulsion to achieve the highest speed possible within such a space, but the maximum velocity is much lower than the maximum velocity of the first ship. In other words, the speed of light through vacuum is lowered within the trail of lightspeed ships.” “How much lower?” “Theoretically, it could be reduced to zero, but that’s not achievable in reality. But if you adjust the curvature ratio of Halo’s engine to the maximum, you can lower the speed of light in its trail down to exactly what we’ve been looking for: sixteen point seven kilometers per second.” “Then you’d have...” AA said, staring at Luo Ji. The black domain, Cheng Xin thought.

Highlight (Yellow) | Page 534

“The black domain,” Luo Ji said. “Of course, a single ship is insufficient to produce a black domain containing an entire star and its planetary system. We calculated that it would take more than a thousand curvature propulsion ships to accomplish such a thing. If all these ships started near the Sun and spread out in every direction at lightspeed, the trails they produced would expand and connect to each other, forming a sphere that

contained the entire Solar System. The speed of light within this sphere would be sixteen point seven kilometers per second—a reduced-lightspeed black hole, or a black domain.”

Highlight (Yellow) | Page 535

Cheng Xin said what he couldn't bring himself to say. “There were thirty-five years between the Halo City Incident and the completion of the Mercury base. Thirty-five precious years were lost.” Luo Ji nodded. Cheng Xin thought the way he looked at her was no longer kind, but rather resembled the fires of the Last Judgment. His gaze seemed to say, Child, look at what you've done. Cheng Xin now understood that of the three paths of survival presented to humanity—the Bunker Project, the Black Domain Plan, and lightspeed ships—only lightspeed ships were the right choice. Yun Tianming had pointed this out, but she had blocked it. If she hadn't stopped Wade, Halo City might have achieved independence. Even if the independence was short-lived, they could have discovered the effects of lightspeed trails and changed the government's attitude toward lightspeed ships. Humanity might have had time to construct a thousand lightspeed ships and build the black domain, to avoid this dimensional strike. Humanity could have divided into two parts: those who wanted to fly to the stars, and those who wanted to stay within the black domain and live in tranquility. Each would have gotten what they wanted. In the end, she had committed another grave error. Twice, she had been placed in a position of authority second only to God, and both times she had pushed the world into the abyss in the name of love. This time, no one could fix her mistake for her.

Highlight (Yellow) | Page 537

The Doppler effect of Halo's speed was now visible. The light from the stars ahead shifted to bluish, while the light from the stars behind shifted to reddish. The color shift was apparent in the two-dimensional Solar System.

Highlight (Yellow) | Page 540

The universe shuddered and transformed. All the stars in front of them shot straight ahead, as though that half of the universe had transformed into a black bowl and all the stars were falling into the bottom. They clustered ahead of the ship and fused into a single glow, like a giant sapphire in which it was not possible to distinguish individual stars. From time to time, individual stars shot out of the sapphire and swept past the pure black space to fall behind the ship, changing color the whole way: from blue to green, then yellow, and turning red once they were behind the ship. Looking back from the ship, the two-dimensional Solar System and the stars fused into a red ball like a campfire at the end of the universe. Halo flew at the speed of light toward the star that Yun Tianming had given Cheng Xin.

Highlight (Yellow) | Page 543

Do you think if you stand on your hands, you've lifted the Earth?

Highlight (Yellow) | Page 543

Cheng Xin's only thought was to find Yun Tianming and tell him everything. In her mind, a journey of 287 light-years would take a long time, but the ship's AI informed her that the trip would only take fifty-two hours in the ship's frame of reference.

Highlight (Yellow) | Page 544

Cheng Xin spent a long time gazing out of the portholes at space. She understood that each time a star leapt out of the blue cluster in front, swept past the ship, and joined the red cluster behind the ship, it meant that Halo had passed it. She counted the stars and watched as they turned from blue to red—the sight was hypnotic. Eventually, she fell asleep.

Highlight (Yellow) | Page 544

“We’re now two hundred eighty-six point five light-years from the Solar System,” said the ship’s AI. “So two hundred eighty-six years has already passed back there?” AA asked. She looked as if she had just awakened from a dream. “Yes, if you are using their frame of reference.” Cheng Xin sighed.

Highlight (Yellow) | Page 545

DX3906 had two planets, both of them solid. The one farther from the star was about the size of Mars, but it had no atmosphere and appeared gray in color—so Cheng Xin and AA decided to call it Planet Gray. The other planet, closer to the star, was about the size of the Earth, and its surface resembled the Earth’s: an atmosphere containing oxygen and many signs of life, but without evidence of agriculture or industry. Since it was blue, like the Earth, they decided to call it Planet Blue. AA was very happy that her research had been confirmed. More than four hundred years ago, she had discovered the star’s planetary system. Before then, people had thought it was a bare star without any planets. Through that work, AA had gotten to know Cheng Xin. Without that coincidence, her life would have turned out completely differently. Fate was such an odd thing: Four centuries ago, when she had gazed at this distant world through the telescope, she could never have imagined that she’d come here one day.

Highlight (Yellow) | Page 548

“Where are Blue Space and Gravity now?” Cheng Xin struggled to stand, pulling herself up by the railing of the stairs. Yifan continued down with AA. “In museums.” “Where are the museums?” AA asked. She put her arm around Yifan’s shoulder so that Yifan was practically carrying her down. “On World I and World IV.” “How many worlds are there?” “Four. And two more are being opened up for settlement.” “Where are all these worlds?” Guan Yifan gently deposited AA on the ground and laughed. “A word of advice: In the future, no matter who you meet—human or otherwise—don’t ask for the location of their worlds. That’s a basic bit of manners in the cosmos—like how it’s impolite to ask a lady’s age.... Nonetheless, let me ask you, how old are you now?” “We’re as old as we look,” AA said, and sat down on the grass. “She’s seven hundred and I’m five hundred.”

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Yifan told them that the blue grass really did move. The roots of the grass were also feet, and as the seasons changed, the grass migrated across the latitudes, mainly at night. As soon as AA heard that, she tossed away the stalks of grass she was playing with in her hands. Yifan explained that the blades of grass really were plants, and relied on photosynthesis, possessing only a basic sense of touch. The other plants in this world were also capable

of moving. He pointed to the mountains and they saw the forests moving in the starlight. The trees moved far faster than the grass, and resembled armies marching at night.

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“No, you won’t be able to see anything.” Yifan shook his head and pointed at that patch of sky again. “Even if you go back there now, you wouldn’t be able to see anything. That part of space is empty. The two-dimensional Sun and planets you saw were actually just the result of the release of energy when three-dimensional material collapsed into two dimensions. What you saw wasn’t two-dimensional material, only the refraction of electromagnetic radiation at the interface between two-dimensional and three-dimensional space. After the energy was released, nothing would be visible. The two-dimensional Solar Space has no contact with three-dimensional space.” “How can that be?” Cheng Xin asked. “It’s possible to see the three-dimensional world from four-dimensional space.” “True. I personally got to see three-dimensional space from four-dimensional space, but it’s not possible to see the two-dimensional world from three dimensions. This is because three-dimensional space has thickness, meaning that there is a dimension that could stop and scatter the light from four-dimensional space, making it visible from four dimensions. But two-dimensional space has no thickness, so light from three-dimensional space passes through it without hindrance. The two-dimensional world is completely transparent and cannot be seen.” “There’s no way at all?” AA asked. “No. In theory, nothing allows it.”

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“There’s only one way to detect the presence of the two-dimensional Solar System from three-dimensional space: gravity. The gravity of the Solar System still has an effect, so, in that empty space ought to be detectable as an invisible source of gravity.” Cheng Xin and AA looked at each other thoughtfully. “Sounds like dark matter, doesn’t it?” Yifan laughed. Then he changed the subject. “Why don’t we talk about the date you came for?” “You know Yun Tianming?” AA asked.

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But she noticed that while he discussed Worlds I, II, and IV, nothing was said about III. “I’ve never been there. No one else has, either. Well, it’s more accurate to say that anyone who goes there cannot return. That world is sealed inside a light tomb.” “A light tomb?” “It’s a reduced-lightspeed black hole produced by the trails of lightspeed ships. Something happened on World III that caused them to think that their coordinates had been exposed. They had no choice but to turn their world into such a black hole.” “We call such a place a black domain.” “Ah, good name. As a matter of fact, the people of World III initially called it a light curtain, but outsiders referred to it as a light tomb.” “Like a shroud?”¹⁰ “That’s right. Different people see things differently. The inhabitants of World III said it was a happy paradise—though I don’t know if they still think that way. After the light tomb was completed, it was impossible for any message from that world to reach the outside. But I think people there are pretty happy. For some people, safety is the sine qua non for happiness.”

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Yifan said, “Lightspeed spaceflight was a tremendous milestone for humankind. It was like another Enlightenment, another Renaissance. Lightspeed flight fundamentally transformed human thinking and changed

civilization and culture.” “I can see that. The moment I entered lightspeed, I felt myself change. I realized that I could, in my lifetime, leap across space-time and reach the edge of the cosmos and the end of the universe. Things that used to seem only philosophical suddenly became concrete and practical.” “Yes. Things like the fate and goal of the universe used to be only ethereal concerns of philosophers, but now every ordinary person must consider them.” “Has anyone in the new world thought of going to the end of the universe?” “Of course. Five ultimate spaceships have already been launched.” “Ultimate spaceships?” “Some call them doomsday ships. These lightspeed ships have no destination at all. They turn their curvature engines to maximum and accelerate like crazy, infinitely approaching the speed of light. Their goal is to leap across time using relativity until they reach the heat death of the universe. By their calculations, ten years within their frame of reference would equal fifty billion years in ours.

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Yifan asked, “Why don’t you tell me what the most powerful weapon for a civilization possessing almost infinite technological prowess is? Don’t think of this as a technical question. Think philosophy.” Cheng Xin pondered for a while and then struggled to shake her head. “I don’t know.” “Your experiences should give you a hint.” What had she experienced? She had seen how a cruel attacker could lower the dimensions of space by one and destroy a solar system. What are dimensions? “The universal laws of physics,” Cheng Xin said. “That’s right. The universal laws of physics are the most terrifying weapons, and also the most effective defenses.
